



CEE2ACT

**Empowering the Central and
Eastern European Countries to Develop
Bioeconomy Strategies and Action Plans**

**Procedure for the creation of multi-stakeholder working
groups and platforms**



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EXECUTIVE SUMMARY

This report aims to explore the foundations, key considerations, and critical steps involved in developing a National Bioeconomy Strategy, with a focus on the CEE2ACT member countries. Drawing from the key insights from other European countries that has adopted and introduced the Bioeconomy Strategy within their countries, as well as initial reports presenting the baseline assessment and stakeholder engagement of the CEE2ACT member states, we summarized critical lessons for the development of the Strategy. Opportunities and potential barriers, as well as expectations of the stakeholders are also gathered from the CEE2ACT national workshop.

The National Bioeconomy Strategy is thoughtfully developed by carefully considering an overview of each country's unique situation and incorporating input from sector-specific divisions. The overarching vision is then expanded upon and translated into well-defined objectives.

A questionnaire in the form of an MS Excel spreadsheet was created to obtain the input data from all CEE2ACT member states. The visions and goals were to be filled in and divided into four pre-determined categories. These dimensions are based on the formulation of the CEE2ACT project and are also in line with the sustainable development objectives, as follows (1) bioeconomy strategies for research and innovation, (2) social challenges, (3) finance and investment, and (4) environment. The collected data underwent a quantitative analysis (frequency analysis) commonly employed in the social sciences, aiming to identify and quantify the words or terms that occur most frequently in the text.

We identify several shared topics that serve as key elements for enhancing the implementation of these strategies, when comparing the thematic groups within the submitted visions and objectives of bioeconomy strategies. These common themes include collaboration, education, and economic considerations.

Collaboration is essential across the three selected pillars of bioeconomy strategies: research and innovation, addressing social challenges, and managing finance and investment. In this context, it is broadly defined to encompass international, inter-ministerial, inter-company, public-private, and science-education-business collaborations, as well as fundraising initiatives. In the context of the same three pillars (research and innovation, social challenges, finance and investment), the education sector holds significance, encompassing both formal education (general and specialized) and informal education through public awareness initiatives.

Economic aspects entail alignment with economic concepts and, with some degree of simplification, aid in comprehending the role of bioeconomy in society. The core idea is that investments in bioeconomy development should yield multiple returns in the form of potential income. Another set of themes revolves around the management of renewable resources, appearing in two pillars—research and innovation and the environment. This reflects the functional realization of economic and environmental objectives within the framework of sustainable development.

DISCLAIMER

Funded by the European Union. Views and opinions expressed are however those of the author(s) only and do not necessarily reflect those of the European Union or the European Research Executive Agency (REA). Neither the European Union nor the granting authority can be held responsible for them.

1. INTRODUCTION

In an era marked by increasing environmental concerns, resource constraints, and the pursuit of sustainable development, the bioeconomy, a dynamic and multifaceted concept, offers a transformative approach to achieve economic growth, environmental sustainability, and societal well-being. This report delves into the vital process of crafting a National Bioeconomy Strategy for CEE2ACT countries, a pivotal framework designed to channel the potential of biological resources into a sustainable, regenerative, and economically viable future.

As we navigate the current complexities, a comprehensive and well-informed national bioeconomy strategy is apparent. It presents a unique opportunity to align their economic aspirations with ecological imperatives, fostering innovation, job creation, and resilience while concurrently addressing ecological challenges and securing essential resources for the future. This report serves as a roadmap for understanding the vision and fundamental principles, challenges, and best practices in the development of a National Bioeconomy Strategy within the CEE2ACT countries, providing insights into how this strategic framework can serve as a catalyst for prosperity, environmental stewardship, and societal advancement.

The report aims to explore the foundations, key considerations, and critical steps involved in developing a National Bioeconomy Strategy, with a focus on the CEE2ACT member states.

Visions and strategies play crucial roles in long-term strategic planning. In the context of general strategic management, a vision is a guiding, inspirational description of a country's envisioned evolution in its bioeconomy over the long term. Meanwhile, the goals are specific, measurable, achievable, relevant, and time-bound steps aimed at realizing the stated vision.

The previous stages of the projects resulted to outcomes in other work packages. , In this case, for example, WP2 focused primarily on describing the current state of bioeconomy predispositions, practices, and barriers in partner countries. In WP6, we seek insights into future visions, plans, and long-term perspectives.

2. METHODS

This deliverable is intended to provide a guidance for developing the national bioeconomy strategy in countries that have not yet adopted such a strategy. The initial step in the methodology involves reviewing the experiences from distinctive countries in developing their national bioeconomy strategies. The D5.1 report, encompassing both the positive aspects and drawbacks during the strategy development process, can be used as lessons learned by CEE2ACT countries.

The second step is initiation of preliminary bioeconomy, which will be based on a sectoral basis, i.e., research and innovation, society, finance and investments, as well as the environment, or it can be directly implemented at the national level (as illustrated in Figure 1). In this step, an analysis of the present situation, obstacles, and opportunities, from the D2.1 (baseline assessment report), which examines the implementation of bioeconomy and policy development in all CEE2ACT target countries, has been completed and condensed. Drawing from the findings outlined in D2.1, this report outlines the initial conditions for each individual CEE2ACT target country.

Regarding the societal sector, additional elaboration has been provided on the report stemming from the D3.1 Stakeholder Engagement Plan. Additionally, the findings from the national workshop are also summarized to support the development of the strategy. Since at any stage of the development, the active involvement of the CEE2ACT Bioeconomy Hub is anticipated to support the key stakeholders (the government). From practical implementation inspiration to assistance in drafting the bioeconomy strategy that includes vision and objectives.

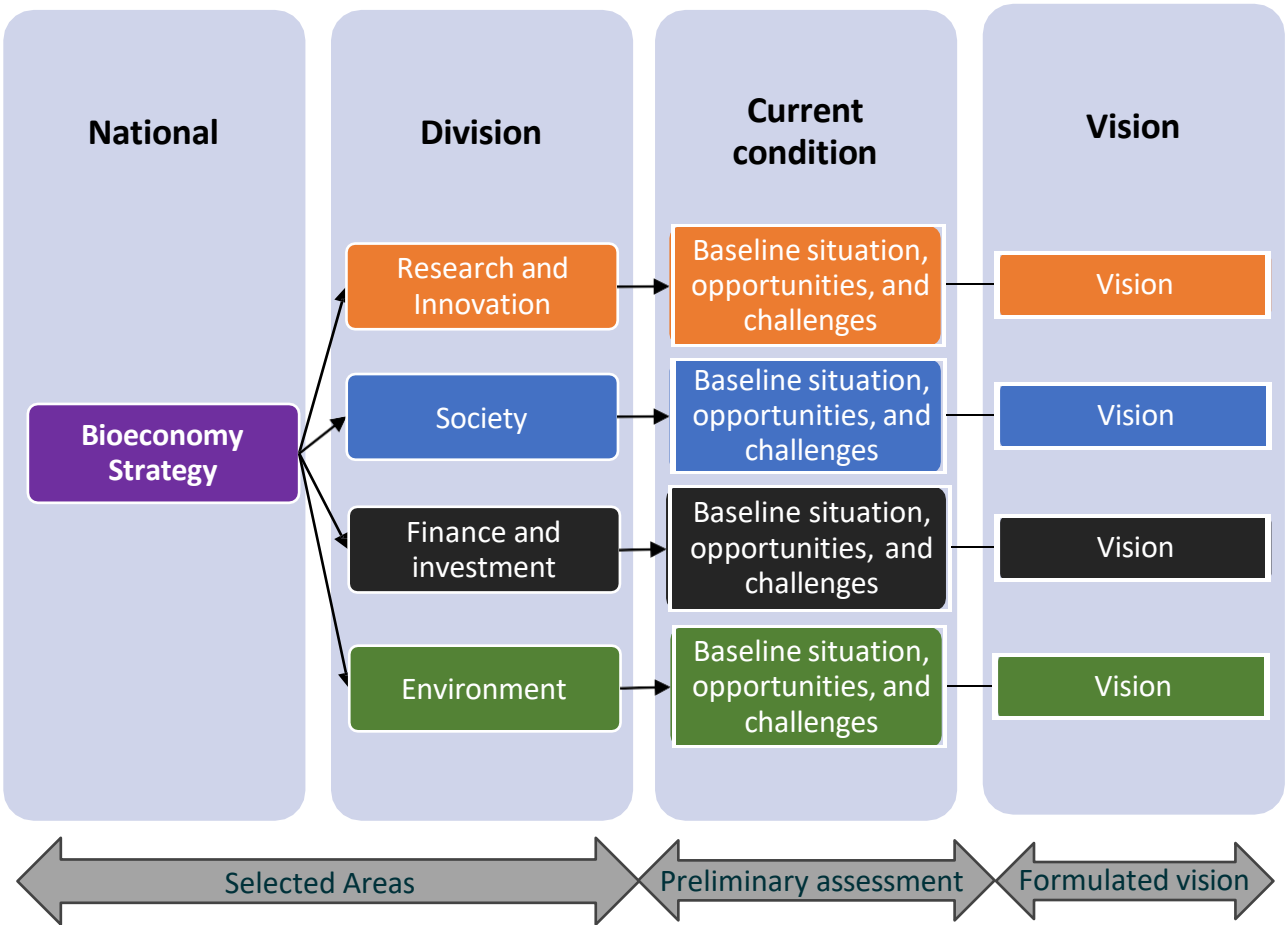


Figure 1. Framework of the guidelines for drafting the Bioeconomy strategy at national level and/or main sectors

We anticipate a crucial distinction between the current state description and the envisioned future. Hence, a questionnaire in the format of an MS Excel spreadsheet (refer to Table 1) was developed for collecting input data and was sent to all partners for filling visions and objectives. The vision and goals were designed to be entered and categorized into four pre-defined dimensions. These dimensions align with the CEE2act project's rationale, aims, and roadmap, and are consistent with the objectives of sustainable development, including an environmental dimension, which we incorporated. The four pre-defined dimensions are:

- 1. bioeconomy strategies for research and innovation
- 2. social challenges, considering gender aspects and re-skilling needs
- 3. finance and investment
- 4. environment

Table 1. Visions and objectives of the national bioeconomy (WP6.1)

1. bioeconomy strategies for research and innovation		2. social challenges, considering gender aspects and re-skilling needs		3. finance and investment		4. environment	
vision	objective(s)	vision	objective(s)	vision	objective(s)	vision	objective(s)

We also draw from international or regional experiences, case studies, and lessons learned to inform and inspire the creation of tailored strategies that reflect the unique characteristics and aspirations of individual country. The goal is to equip policymakers, stakeholders, and citizens within the CEE2ACT countries with the knowledge and tools needed to chart a sustainable and affluent course through the intricate landscape of the bioeconomy. As countries increasingly recognize the multifaceted benefits of the bioeconomy, the creation of individualized bioeconomy strategies becomes an imperative. These strategies serve as blueprints for leveraging biological resources, fostering innovation, and driving economic growth while ensuring ecological sustainability and societal well-being. Furthermore, target countries developed a future vision and established goals until 2030 within the selected areas using results from Table 1.

The data underwent a quantitative analysis loosely inspired by frequency analysis commonly employed in the social sciences. This method involves quantitatively analyzing text content by monitoring the frequency of selected words or terms in a text corpus aiming to identify and quantify the words or terms that occur most frequently in the text. The method was adapted to the data sent by each project partner see the filled tables in Annex1.

This method facilitated the standardization of input formats for analysis, thus, enabling subsequent generalizations. During the frequency analysis of visions and goals, recurrent sub-terms, presented in various forms or assigned high priority, were identified. These sub-terms were then grouped into subcategories based on similarity or thematic relatedness. The outcome was the identification of key themes prioritized within the categories, along with the recognition of cross-cutting themes that influence multiple categories and may have a unifying role.

3. KEY LESSONS GAINED FROM THE CONTRIBUTING COUNTRIES IN BIOECONOMY STRATEGY DEVELOPMENT

3.1. Austria

The Austria's bioeconomy¹ intends to achieve an economy that combines technology and ecology, emphasizing the use of renewable resources, particularly from forests and agriculture, while reducing dependence on fossil raw material imports.

The development of the strategy involved collaboration between several federal ministries. The federal government received guidance from a high-level committee of experts to shape the strategy. An extensive consultation process was also conducted to ensure input from all relevant stakeholders.

Following the introduction of the Strategy, multiple open discussions from all potential stakeholders were done, and clearly showed the great interest and the opportunities and challenges associated with the topic in Austria. Options for action are developed in the bioeconomy strategy and then supplemented with concrete measures in an action plan². This built on Austria's strengths in research, agriculture and forestry and opened up new areas.

3.2. Belgium

Belgium conducted an analysis of the initial conditions in the realms of society, economy, and the environment, in addition to assessing the potential biomass resources as foundational elements for their bioeconomy endeavors. The nation has accumulated substantial experience from collaborations with other countries, which are pivotal in achieving the essential partnerships and collaborations among key stakeholders. At the same time, the main challenge in policy coherence, insufficient coordinated regulations and policy is recognized. This analysis was undertaken to reinforce the Flemish bioeconomy³ vision and to formulate the strategies required for its realization.

¹ <https://www.bmk.gv.at/en/topics/climate-environment/climate-protection/bioeconomy/strategy.html>

² https://www.bmk.gv.at/themen/klima_umwelt/klimaschutz/biooekonomie/aktionsplan.html

³ <https://publicaties.vlaanderen.be/view-file/13902>

3.3. Finland

The Finnish bioeconomy⁴ primarily revolves around forestry, related industries, construction, energy, agriculture, and the food sector, all of which are integral components. Additionally, the country's focus is in producing high-value added commodities. For the bioeconomy, numerical targets and associated indicators have been set. The challenges faced in the country's bioeconomy development include the limited involvement of the average citizen in the process, and the predominance of forest-based industries, which could potentially hinder innovation opportunities from other sectors.

3.4. Germany

The German bioeconomy⁵ is oriented towards addressing societal needs in a sustainable manner and establishing a strong connection between the economy and ecology. The initial approach to bioeconomy development in Germany was rooted in the necessity to foster innovation through research and development. Two foundational elements of the German bioeconomy are agriculture and forestry, and critical knowledge domains include biofuels and biorefineries, although numerous other sectors contribute to the development and production of bio-based products in Germany. Effective collaboration, both at the national and international levels, is a crucial element in advancing the country's bioeconomy. Germany also commits to invest on the funding of research and innovation in bioeconomy. However, similar to Finland, it is acknowledged that there are challenges related to the underrepresentation of certain stakeholders in the process that could limit the actions and implementation.

3.5. Spain

The initial phase of Spain's bioeconomy journey begins with a political commitment that provides support and a vision for this initiative. This is followed by the formation of a working group consisting of diverse stakeholders from both the public and private sectors, tasked with drafting the working document. Following a public consultation, the finalized Spain Bioeconomy strategy⁶ was disseminated through emails and social networks. Spain also conducted an assessment of the potential biomass to be prioritized. In spite of this, the country acknowledges the need for greater stakeholder representation, like Germany and Finland.

3.6. Sweden

The development of Sweden's bioeconomy strategy included with a focus on research and innovation in 2012⁷, emphasizing the priority of achieving fossil-free

⁴ https://www.biotalous.fi/wp-content/uploads/2022/05/The-Finnish-Bioeconomy-Strategy-Sustainably-towards-higher-value-added-VN_2022_5.pdf

⁵ https://www.bmel.de/SharedDocs/Downloads/EN/Publications/national-bioeconomy-strategy.pdf?__blob=publicationFile&v=2

⁶ <https://bioeconomia.chil.me/download-doc/102159>

⁷ https://www.formas.se/download/18.462d60ec167c69393b91e60f/1549956092919/Strategy_Biobased_Ekonomi_hela.pdf

competitiveness in bioenergy and bio-based feedstock for industries⁸. This was followed by a dedicated effort towards circular economy promotion starting in 2018⁹. The Swedish development depicts the adoption process that initiated from division-specific level into national bioeconomy strategy. A comprehensive national bioeconomy strategy is currently in the works and is expected to be finalized by late 2023. Integrated biorefineries and biorefinery technologies, which encompass aspects of energy efficiency and environmental technology is an important step in the bioeconomy development in the country. Furthermore, Sweden has valuable insights stemming from private sector engagement in bioeconomy strategy development and various collaborative efforts involving research, industry, and government within national strategic innovation programs and regional innovation hubs. Based on these experiences, one of the challenges is ensuring the preparation of a sustainable biomass supply to meet the domestic demand.

3.7. The Netherlands

The Netherlands introduced its bioeconomy strategy in 2018¹⁰, with a strong emphasis on key knowledge and business domains. These include agrifood, biofuels, green chemistry, biomaterials, and biorefineries.

The Netherlands has demonstrated proficiency and expertise in establishing collaborative networks involving multiple stakeholders. Notably, a 'quadruple helix collaboration' model has been employed, encompassing government, academia, industry, and civil society. This approach fosters learning and alignment among the various actors involved in bioeconomy development endeavors.

However, challenges encountered in Dutch bioeconomy strategy development include the difficulty of creating social awareness regarding planetary boundaries, as well as a more general societal and political indifference towards recognizing the urgency of sustainability transitions.

⁸ https://fossilfrittssverige.se/wp-content/uploads/2021/11/Biostrategi_ENG.pdf

⁹ <https://www.delegationcirkularekonomi.se/om-oss>

¹⁰ http://www.bio-economie.nl/wp-content/uploads/2019/12/Min-Econ-Zaken_2018_De-positie-van-de-bio-economie-in-Nederland_brochure.pdf

4. OVERVIEW OF THE INITIAL STATUS IN CEE2ACT TARGET COUNTRIES

4.1. Bulgaria

4.1.1. Research and Innovation

Research, Development, and Innovation (RDI) initiatives include Bioeconomy: The Sustainable Economy of the Future, a study of the potential of bioeconomy in Bulgaria and a study of the European success examples in a bioeconomy (Agriculture University Plovdiv), analysis and profile of the state and potential of a regional bioeconomy. Primary solid fuel is the main form of renewable energy, constituting about 70% and hydro about 10% in the national renewable energy mix. The other sectors, such as solar and wind energy and other less-developed renewable energy technologies, require research and innovation in those areas. There is a low share of renewable energy in the transport of 7%, which shows much reliance on non-renewables in transportation.

The share of renewables in electricity is 19% below the average of all target countries (31%). The share of renewables for heating and cooling constitutes 30%, the same as the average of all target countries but below Sweden (66%) and Finland (55%). Bulgaria has also instituted various initiatives in agriculture, forestry, organic waste, food, wood and biobased chemicals. This shows the potential for bioeconomy through research development in these areas, including research on further developments in engines and fuels that can support renewable energy.

4.1.2. Society

Bulgaria has developed a wide range of strategies that could help transition the country to a bioeconomy. These include a Strategic plan for the development of agriculture and rural areas 2023- 2027, a national strategy for the development of the forestry sector for the period up to 2030, an integrated plan for energy and climate of the Republic of Bulgaria 2021 – 2030, Strategy for the transition to a circular economy for the period 2022-2027, and Renewable Energy Act, 2011. There are also collaborations to support the development of bioeconomy, including the implementation of activities such as meetings at the Agricultural University Plovdiv Regional Hub for bioeconomy, Mission Green Bulgaria, and the Annual Working Meeting of the National Scientific Program “Healthy Foods for a Strong Bioeconomy and Quality of Life” (NNP-FOOD).

Bioeconomy Working Group (National Bulgarian Hub) workshop was held to engage stakeholders to join efforts to prepare a vision and to identify the strategic goals and priorities for the development of a National bioeconomy Strategy and an Action plan. The workshop was held with the support of the Ministry of Agriculture under the CEE2ACT project and Horizon Europe of the European Union. Stakeholders were drawn from the Forestry Executive Agency, the Bulgarian Chamber of Commerce, forest owners, universities, and other institutions. The workshop acknowledged the roles of other key stakeholders such as the Ministry of Economy, the Ministry of Innovation and

Growth, the Ministry of Energy, Ministry of Environment and Water, in the development of the bioeconomy in Bulgaria, despite their absence. For future task of the crafting the national bioeconomy strategy, all key Ministries will be involved.

Nevertheless, the nation lacks national strategy documents and a coherent bioeconomy policy. Although there are distinct sectoral and regional strategies, they don't help solve the issue as a whole and don't provide enough resources, technology, or ecological support for the bioeconomy's growth. The Agricultural Academy created a draft Strategy for the Development of the Bioeconomy 2023–2030 as part of the BIOEASTsUP initiative in order to facilitate public discussion and the adoption of strategic documents. Infrastructure issues and a lack of cooperation among stakeholders are obstacles to the bioeconomy's implementation.

4.1.3. Finance and Investment

GDP per capita of Bulgaria in 2019 was approximately 8,460 Euros (based on calculation from normalized to NL, which has the highest GDP with EUR 47,000 per capita).

The most significant contribution to the bioeconomy sector is Food, Beverage, and Tobacco ($\pm 47\%$), higher than the average contribution of 41% across all CEE2ACT countries, including the Baltic States. This was followed by agriculture ($\pm 29\%$), which is also higher than the average contributions of all target countries (21%). The contribution of the other equally essential sectors is comparatively low regarding their bioeconomy turnover. The country has also instituted various initiatives in different sectors including agriculture (6), forestry (4), organic waste (4), food (6), wood (3), biobased chemicals (3), bioenergy (3) and biotechnology (1). This is an excellent opportunity to sustain the bioeconomic industry by providing raw materials or biowaste. Some challenges include financial barriers, lack of collaboration, and lack of infrastructure and capacities that hinder the development of bioeconomy.

4.1.4. Environment

About 84% of biomass is obtained from agriculture and 16% from forestry. Over 70% of biomass is consumed for energy and more than 20% for different materials. This makes Bulgaria one of the highest in terms of biomass consumption for energy among the target countries. This could contribute to transforming into a fossil-free energy sector to help mitigate climate change. The country has over 30% arable land and over 30% forest land area, constituting the primary sectors of the bioeconomy. Primary solid fuel is the main form of renewable energy, constituting about 70% and hydro about 10% in the renewable energy mix. About half of the total biowaste generated per capita is not recovered, showing low recovery rates and the potential for waste utilization in a bioeconomy. The separate collection rate of biowaste is below the European average of about 50%, showing a high potential for biowaste collection and the need to improve the collection rate to sustain the bioeconomy. The country has about 2.5% share of organic farming in the total agricultural land area, which is low. Organic farming has a lower burden on CO₂ production due to the build-up of soil carbon and the decreased use of inorganic fertilizers that release nitrous oxides – a potent greenhouse gas causing climate change. Indicators of environmental health: Biochemical Oxygen Demand (BOD) in rivers (2.28mgO₂/l), phosphate in rivers (0.1mgPO₄/l), nitrate in groundwater (29.79mgNO₃/l). The BOD shows moderate pollution levels, and the phosphate and

nitrate levels are within acceptable limits. This reflects good environmental health and should be safeguarded.

4.2. Croatia

4.2.1. Research and Innovation

Primary solid fuel is the main form of renewable energy in the renewable energy mix, constituting about 60% and hydro over 20%. There is a need for Research and Innovation in using solar and wind energy and other less-developed renewable energy technologies. There is a low share of renewable energy in transport of 1%, which creates room for research and development to increase their contributions. This also depicts the high dependence on combustion engines and the electrification of the transport sector, and the use of synthetic fuels. The share of renewables in electricity is 46%, above the average of all target countries (31%) but lower than the average (50%) of Austria, Sweden, Croatia, Romania, and Latvia. The share of renewables for heating and cooling constitutes 37%, which is above the average of all target countries (30%) but below Sweden (66%) and Finland (55%). This implies the much potential for research development in these areas, including research on further developments on engines and fuels for renewable energy.

4.2.2. Society

Strategies implemented in Croatia related to bioeconomy include the Common Agricultural Policy (CAP) Strategic Plan, National Law on Forests, circular economy (Waste Management Plan of the Republic of Croatia for the period 2017–2022 (OG 3/17, 1/22), Integrated National Energy and Climate Plan. However, there is no dedicated strategy for a circular economy, even though a Waste Management Plan exists.

The role of the newly established National Hub is to support in the development and implementation of the Bioeconomy Strategy, of which the members were drawn from various representative sectors (from bio-based value chain actors, investors, SMEs, research institutions, academia, environmental organizations, NGOs and CSOs, industrial chambers, textile and pharmaceutical industries, policy and decision-makers, and public administrators).

4.2.3. Finance and Investment

The GDP per capita of Croatia in 2019 was approximately 5,640 Euros (based on calculation from normalized to NL, which has the highest GDP with EUR 47,000 per capita). Food, beverage, and tobacco products constitute the largest portion of the bioeconomy sector, accounting for around ($\pm 48\%$), which is more than the average of 41% for all CEE2ACT nations, including the Baltic States. Agriculture contributes $\pm 23\%$ to the bioeconomy, which is higher than the 21% average contribution from all target countries. The contribution of wood, wood products, and furniture to the bioeconomy sector is approximately 12%. The other equally important sectors contribute relatively little to the bioeconomy. This is an opportunity to support the bioeconomic sector by supplying biowaste or raw materials.

4.2.4. Environment

Approximately 33% of biomass comes from forestry, 1% from fishing, and 66% comes from agriculture. Less than 10% of biomass is used for other purposes and more than 90% is used for energy. Croatia now ranks among the target countries with the largest energy consumption from biomass. This may help the energy sector transition to one free of fossil fuels and slow down climate change. The principal sectors of the bioeconomy in the country are comprised of around 30% forest land and 15% arable land. The extent of land designated as other land is more than 50%. In the renewable energy mix, primary solid fuel accounts for roughly 60% of the total energy, with hydropower accounting for the remaining 20%. Out of the approximately 90 kg of waste produced per capita, less than 30 kg are collected by households, and low recovery rates are evident in the fact that almost half of the total waste (including that from industries and commercial activities) is not recovered. This demonstrates a great potential for bioeconomy in the waste management sector.

The separate collection rate of biowaste is below the European average of about 50%, showing a high potential for biowaste collection and the need to improve the collection rate to sustain the bioeconomy. There is a need to explore other biowaste products apart from animal by-products to sustain anaerobic treatment capacities. The country has about 7% share of organic farming in the total agricultural land area. Organic farming has a lower burden on CO₂ production due to the build-up of soil carbon and decreased use of inorganic fertilizers that release nitrous oxides – a potent greenhouse gas causing climate change. Indicators of environmental health, BOD in rivers (1.85mgO₂/l), phosphate in rivers (0.03mgPO₄/l). The BOD shows that the rivers are clean and devoid of pollution, and the phosphate is also within acceptable limits. This shows good environmental health and should be safeguarded. There is no data for nitrates in groundwater.

4.3. Czech Republic

4.3.1. Research and Innovation

In Research and Innovation of biomass production, agriculture potentially contributed to more than 60%, followed by forestry (about 35%). The low proportion of renewable transport utilization (7%) and electricity (14%) implies the opportunity for research and development in these areas. The implementation of several initiatives in different sectors, including agriculture, forestry, fisheries, aquaculture, organic waste, and among others, requires research and development to improve those sectors.

4.3.2. Society

Roadmaps for the transition to a Circular Economy have been implemented in the Czech Republic. Sectors that are the backbones of the Bioeconomy have acknowledged the Strategy, for instance the Agriculture Policy and the Forest Policy. The country also implements the National plan of the Czech Republic in the field of energy and climate and Environmental Policy. At the same time, percentage of the Czech society used renewable transport, electricity, heating, and cooling is still low, hence, promotion of

the utilization can be intensified to prepare the transition into the implementation of the Bioeconomy Strategy in the future.

Involving various ministries is crucial for success, and there's already strong interest from multiple ministries. Additionally, umbrella organizations such as clusters and associations, in close collaboration with these ministries, will engage with a wide range of stakeholders involved in bioeconomy-related matters. It's important to note that this approach is inclusive and welcomes stakeholders beyond these two groups. The established BIOHUB CZ is already working with multiple ministries, bridging the gap between them. Typically, ministries have a good understanding of bioeconomy topics, and they aim to fill knowledge gaps in the private sector. Two national initiatives (Association and Platform) have been developed in the country, consisting of various bioeconomy-related areas, such as agriculture, forestry, fisheries, aquaculture, wood products, bioenergy, etc. Challenges to implementation of bioeconomy include lack of motivation and legal barriers.

The Ministries, e.g., Ministry of Agriculture, and Ministry of the Environment are the key stakeholders in developing the national bioeconomy strategy. In this sense, the Czech Bioeconomy Hub from a wide range of stakeholders from the ministries, companies, interested associations, NGOs, and universities, some of which include BIOEAST HUB CZ, Bioeconomy Platform of the Czech Republic, Czech Ministry of Agriculture, and Ministry of the Environment will support the development of national bioeconomy strategy through providing ideas and knowledge, e.g., baseline situation in bioeconomy sectors, best practices, etc.

4.3.3. Finance and investment

GDP per capita of the Czech Republic in 2019 was approximately 21,150 Euros (based on calculation from normalized to NL, which has the highest GDP with EUR 47,000 per capita). The Czech CAP plan has been approved for the implementation of 2023-2027, giving the opportunity to develop the agriculture sector and rural development programs, which include forest-based sector, that is in line with the Bioeconomy Principles.

Food, Beverage, and Tobacco sector has the largest contribution (about 40%) to bioeconomy in the country, followed by agriculture ($\pm 20\%$) and forestry ($\pm 15\%$). The country has also instituted a number of initiatives in different sectors including agriculture (2), forestry (2), fisheries (2), aquaculture (2), organic waste (2), and among others. This serves as a great opportunity for the sustenance of the bioeconomic industry in terms of the provision of raw materials or waste. Challenges to implementation of bioeconomy include financial barriers.

4.3.4. Environment

Over 60% of biomass is obtained from agriculture and 36% from forestry. More than 75% of biomass production is used for energy, and the rest is used for raw materials. Primary solid fuel is the main form of renewable energy in the national renewable energy mix constituting over 60% and more than 10% biogas usage. This makes Czech Republic one of the highest in terms of biomass consumption for energy among the target countries. This could contribute to transforming to a fossil-free energy sector to



help mitigate climate change. On average, bio-waste collection in the Czech Republic closely aligns with the European-wide average (approximately 50%), highlighting the potential for supporting the implementation of environment-related policies. The country has more than 30% arable land and over 30% forest of land area constituting the primary sectors of bioeconomy. The other land areas constitute built areas or unused lands. Approximately 15% of the nation's total agricultural land area is used for organic farming. Because organic farming increases soil carbon and reduces the need of inorganic fertilizers, which generate nitrous oxides, a powerful greenhouse gas that contributes to climate change.

Environmental health indicators include groundwater nitrate (17.98 mgNO₃/l), phosphate (0.14 mgPO₄/l) and BOD (2.61 mgO₂/l) in rivers. Phosphate and nitrate levels are within permissible limits, and the BOD indicates moderate pollution levels. These indicators could be enhanced through cleaner production processes since they signal a healthy or a deteriorating ecosystem. However, over 60% of rivers have a BOD content of 2-4 mgO₂/l or higher, and roughly 34% of rivers have a BOD concentration of less than 1.4-2 mgO₂/l. This demonstrates the necessity of taking action to reduce the amount of nutrients from different production activities, including agriculture, by improving biowaste treatment. One way to achieve this would be to implement circularity or bioeconomy through water-smart solutions that can aid in the prevention of water pollution.

4.4. Greece

4.4.1. Research and Innovation

Greece has a good distribution of renewable energy sources comprising hydropower (about 11%) and solid biomass (25%), solar thermal and solar photovoltaic energy (over 10% each), wind energy (about 20%), and heat pumps (about 10%), pure biodiesel and biogases constitute about 5% each and geothermal in a share of less than 1%. Due to the excessive reliance on combustion engines, the electrification of the transportation industry, and the use of synthetic fuels, the percentage of renewable energy in transportation is low (4%). However, with innovation and research in solar, wind, and other less-developed renewable energy technologies, the less-developed areas could be expanded.

About 24% of electricity comes from renewable sources, less than the average for all target countries (31%). The share of renewables for heating and cooling constitutes about 28% which is below the average of all target countries (30%), Sweden (66%) and Finland (55%). This suggests that there is an opportunity for research and development in these fields, including studies on future advancements in engines and fuels reliant on renewable energy sources.

4.4.2. Society

According to the baseline study, Greece does not have a formal national policy for the bioeconomy. Nonetheless, numerous bioeconomy-related plans/programs have been developed including energy and climate change. Therefore, the country is in the process of transitioning and decarbonizing, as well as addressing its energy concerns. The bioeconomy strategic plans have been delivered, and the next phase will be the

formulation of the national bioeconomy strategy. Some national plans/programs related to bioeconomy include the Strategic Plan of the Common Agricultural Policy of Greece 2023-2027 (CAP), National Forest Strategy, National Biodiversity Strategy & Action Plan, Circular Economy Action Plan, Article 97- End of the Use of plastic bags, Integrated National Energy and Climate Plan, National Climate Change Adaptation Strategy (NCCAS), National Strategy for the Industry and Action Plan, The Rural Development Program (2014-2020). Notwithstanding these efforts, the country is faced with a lack of collaboration among stakeholders and legal barriers that thwart bioeconomic implementation efforts. Currently, there are four initiatives implemented in the various bioeconomy sectors in this country.

The newly established Hub will support the development of national bioeconomy strategy through networking and knowledge sharing, facilitating funding opportunities, access to expertise, and supporting sustainable development and policy Resource Centre. This Hub received official endorsement and support from the Ministry of Environment and Energy.

4.4.3. Finance and Investment

The GDP per capita of Greece in 2019 was approximately 16,920 Euros (based on calculation from normalized to NL, which has the highest GDP with EUR 47,000 per capita). The country produces over 90% of biomass from the agriculture sector alone. The biomass from forestry and fisheries combined constitute less than 10%. The high level of biomass in the agriculture sector is an opportunity for investment in bioeconomy in that area. However, the challenges to the implementation of bioeconomy are mainly financial barriers

Food, beverages, and tobacco sector contributed the highest proportion (about 50%) to the Greek bioeconomy turnover, followed by agriculture (above 35%). Bio-based chemical and materials together with pulp and paper sector, demonstrated an increasing economic contribution to the overall bioeconomy turnover in the country.

4.4.4. Environment

Over 94% of biomass is obtained from agriculture, 5% from forestry, and less than 1% from fisheries. More than 30% of biomass is consumed for energy and more than 60% for different materials. The separate collection rate of biowaste is below the European average of about 50% showing a high potential for biowaste collection and the need to improve the collection rate to sustain the bioeconomy. However, there is a need to increase the treatment capacity for biowaste to accommodate the volume of biowaste generated. The country has about 16% arable land and over 30% forest land area constituting the primary sectors of the bioeconomy. More than 50% of the land area is designated as other lands. Organic farming accounts for around 10% of total agricultural land area in the country. It is an environmentally friendly method of farming that conserves soil carbon and limiting inorganic fertilizers that emit greenhouse gases that impacts the earth's climate system. Regarding environmental health, there is no available data from the baseline study (WP2) that could be used as indicators or to measure the status of environmental health in Greece.

4.5. Hungary

4.5.1. Research and Innovation

In the renewable energy mix, primary solid fuel accounts for more than 60% of the energy and more than 10% of the pure biogasoline. The other renewable energy sources, such wind and solar, which account for 5% or less, are insufficient. There is a low share of renewable energy in transport of 8% requiring some form of research and innovation to expand. This depicts the transportation sector's heavy reliance on fossil fuel engines, the usage of synthetic fuels, and their application. The percentage of renewable energy in electricity is 8%, which is less than the target countries average of 31%.

The share of renewable energy for heating and cooling is 20%, which is lower than the average for all target countries (30%). Several efforts have led to initiatives in areas, including agriculture, organic waste, wood, biotechnology, biobased chemicals, and bioenergy. This means that there is potential for bioeconomy, which might be realized through research and development, including research on future advances on renewable energy engines and fuels.

4.5.2. Society

Several national plans/programs and strategies having links with bioeconomy have been developed. These include the Rural development program focusing on agriculture, the Forest Strategy, the National Circular Economy Strategy, the National Climate Change Strategy, the National Renewable Energy Action Plan, and the National Clean Development Strategy 2020-2050, among others. These strategies have led to the implementation of several initiatives in different sectors including agriculture (1), organic waste (1), wood (1), biotechnology, biobased chemicals (1), and bioenergy (1). However, these initiatives are inadequate in terms of numbers to help expand the bioeconomy. In response to the need to expand the bioeconomy sector, the Ministry of Agriculture, a member of the BIOEAST Initiative, developed and published a strategic bioeconomy concept paper as part of the BIOEASTsUP project (2019-2023). The primary goal of this publication is to serve as a foundation for cross-sectoral and multidisciplinary dialogue on biomass-related sustainability challenges. The concept paper has set in motion some cooperation among stakeholders (e.g., universities, businesses, farmers, investors, representatives from various bio-based industries, etc.). For example, several in-person and online workshops, as well as info-days (such as the CBE JU Infoday on April 26, 2023) have been held. Even with these initiatives, more work has to be done to foster effective collaboration.

Additionally, there is a collaboration between the Prime Minister's Office and Circular Hungary, the organizers of the Circular Economy Technology Platform (CETP). With the methodological assistance of the OECD, this collaboration aims to develop a circular economy (CE) strategy. The CE strategy was introduced to the public in March of 2023 with the main focus on the circularity of biomass besides other waste and plastics. With this level of progress, the development of bioeconomy has reached the stage of capacity building of key stakeholders, created awareness, and enabled the buy-in of new supporters. Regarding the current stage of bioeconomy, plans are at foot to organize more meetings with the involvement of all stakeholders to define the national-level

priorities on bioeconomy. Besides the progress being made, there are challenges to implementation including the lack of awareness, effective collaboration, motivation, and infrastructure, and the lack of specific policy to give a strong backing to bioeconomy.

4.5.3. Finance and Investment

The GDP per capita of Hungary in 2019 was approximately 35,720 Euros (based on calculation from normalized to NL, which has the highest GDP with EUR 47,000 per capita). Food, Beverage, and Tobacco constitute the most significant contribution to the bioeconomy sector at about ($\pm 43\%$), higher than the average contribution of 41% across all CEE2ACT countries, including the Baltic States. The next contributor to the bioeconomy is agriculture ($\pm 32\%$), which is also higher than the average contributions of all target countries of 21%. Wood, wood products, and furniture constitute about 5% of the contribution to the bioeconomy sector. The other equally essential sectors are comparatively low regarding their bioeconomy contribution. The country has also instituted several initiatives in agriculture, organic waste, wood, biotechnology, biobased chemicals, and bioenergy. The number of initiatives per sector is inadequate creating room for investment and expansion. The Challenges to the implementation of bioeconomy include financial barriers and the lack of infrastructure and capacities.

4.5.4. Environment

Agriculture accounts for more than 85% of biomass production, with forestry accounting for roughly 12%. More than 65% of biomass is used for energy, with the remaining 30% used for other purposes. Primary solid fuel accounts for more than 60% of the renewable energy mix, with pure biogasoline (10%), pure biodiesel (5%), solar photovoltaics (4%), geothermal (5%). The renewable municipal waste and the others account for 5% or less. About 50kg/capita of waste is recovered by households out of over 150kg/capita of waste generated by households and two-third of total waste is not recovered, showing low recovery rates. This shows enormous potential for waste management to support the bioeconomy. The separate collection rate of biowaste is below the European average of about 50% showing a high potential for biowaste collection and the need to improve the collection rate to sustain the bioeconomy. However, as biowaste increases with the inclusion of other biowaste in municipal waste, there is a need to increase the installation of treatment infrastructure to accommodate the increase in volumes. The country has more than 40% arable land, over 20% forest land area constituting the primary sectors of the bioeconomy, and 30% of the land area designated as other lands. The country has about 6% share of organic farming in the total agricultural land area. Organic farming has a lower effect on the climate due to ecologically sound practices including minimal application of inorganic fertilizers that contribute to the emission of GHGs causing global warming. Regarding environmental health, there is no available data from the baseline study (WP2) that could be used as indicators or to measure the status of environmental health.

4.6. Poland

4.6.1. Research and Innovation

Research, development, and innovation initiatives highlighted in the baseline survey

include new strategies for bioeconomy in Poland (BioeCon). Primary solid fuel is the main form of renewable energy in the energy mix constituting about 70%, wind about 10%, and pure biodiesel less than 10%. The other renewable energy sectors such as hydro, solar photovoltaic, and solar thermal are inadequately developed. There is a low share of renewable energy in transport of 4% requiring research and development to improve. The low share of renewables in transport is understandable because many transport systems are dependent on fossil fuels for combustion and electricity. The share of renewables in electricity is 13% which is below the average of all target countries (31%). The share of renewables for heating and cooling constitutes 15% which is below the average of all target countries (30%). The potential sectors for bioeconomy research and development include agriculture, forestry, fisheries, aquaculture, organic waste, food, wood, pulp and paper, biotechnology, biobased chemicals, biobased textiles (3), bioenergy, and ecosystem services. This implies there is a wide area of sectors that have potential for bioeconomy research, development. Because of the high dependence on fossil-based transport systems, it is recommended that research and development also be carried out on renewable energy-based engines and fuels to help transition to renewables. Challenges to the implementation of bioeconomy include knowledge barriers and a lack of collaboration among stakeholders.

4.6.2. Society

Poland has many national documents that take into account elements of the bioeconomy, as well as several cross-sectoral initiatives, including bioeconomy cooperation agreements. The Polish Strategic Plan for the Common Agricultural Policy-2023-2027 (CAP), the National Forestry Accounting Plan, the Biodiversity Strategy 2030, the Strategy for Sustainable Rural, Agricultural, and Fisheries Development 2030 (SFDS 2030), the Road Map Transition to Circular Economy, the Integrated National Energy and Climate Plan, and the Renewable Energy Sources Act 2022 are some of these national documents. Furthermore, the country joined the EU's 'Farm to Fork' strategy, which includes the establishment of strategic provisions to support the bioeconomy. These plans and strategies have contributed to the development of several initiatives. Among the target countries, Poland recorded the highest number of implemented initiatives across a wide range of different sectors including agriculture (9), forestry (3), fisheries (5), aquaculture (6), organic waste (8), food (7), wood (3), pulp and paper (3), biotechnology (9), biobased chemicals (6), biobased textiles (3), bioenergy (6) and ecosystem services (3).

This cannot be accomplished without collaboration with ministries, meetings, and conferences to encourage bioeconomy development. The collaboration has resulted in activities such as the Biogas and Biomethane Sector Cooperation Agreement and the Bioeconomy Development Group Initiative. As part of the BIOEASTsUP initiative, Poland has developed a concept paper that could aid in the establishment of a bioeconomy strategy. The concept paper's main focus is on establishing niche bioeconomy sectors such as biogas and biomethane, organic farming (agricultural), aquaculture (inland fisheries), and 'short agri-food value chains'. Inadequate awareness creation and collaboration, a lack of motivation, policy and legal hurdles are among the implementation challenges.

Furthermore, drawing from the results of the stakeholder's workshop, which aimed to



establish a national bioeconomy Hub, participants acknowledged the absence of a clear direction and strategy at the country level. Therefore, a functional Hub can play a vital role in expediting the development of a national bioeconomy strategy. It can achieve this by providing advice to the government and local authorities, generating ideas, and presenting solutions derived from a diverse array of stakeholders.

4.6.3. Finance and Investment

In 2019, Poland's GDP per capita was around 14,100 Euros. Food, beverage, and tobacco contribute the most to the bioeconomy sector, accounting for approximately (53%), which is greater than the average contribution of 41% across all CEE2ACT countries and the Baltic States. Agriculture contributes the most to the bioeconomy (19%), which is lower than the average contribution of all target countries (21%). Wood, wood products, and furniture constitute about 13% (equal to the average of all target countries) of the contribution to the bioeconomy sector. The other equally essential sectors are comparatively low regarding their bioeconomy contribution. The country has a wide bioeconomy sector with investment initiatives in agriculture, forestry, fisheries, aquaculture, organic waste, food, wood, pulp and paper, biotechnology, biobased chemicals, biobased textiles, bioenergy, and ecosystem services. The contribution of these sectors is an opportunity for the industry to generate raw materials or waste to sustain the bioeconomy. Implementation challenges include financial barriers, lack of infrastructure and capacities, and lack of market demand.

4.6.4. Environment

More than 73% of biomass is obtained from agriculture and 27% from forestry. Over 45% of biomass is consumed for energy and more than 50% for different materials. Close to 50% of waste generated is not recovered by households out of the total waste generated. This demonstrates the enormous potential for waste management to help the bioeconomy. The separate collection rate of biowaste is less than 50% of the European average, indicating a strong potential for biowaste collection and the need to enhance the collection rate in order to sustain the bioeconomy. However, when the volume of biowaste increases due to the addition of other biowaste streams to municipal waste, treatment infrastructure capacity must be increased to accommodate the volume. The country has approximately 35% arable land and more than 30% forest land area, which constitute the key sectors of the bioeconomy. Organic farming accounts for only about 3% of total agricultural land area in the country. Organic farming could be encouraged because of its environmentally sound practices that have reduce impact on the climate.

Indicators of environmental health show that the BOD concentration in rivers was 2.95mgO₂/l. This show that the rivers are moderately polluted. However, about 25% of rivers have BOD ranging less than 1.4- 2mgO₂/l and over 70% of rivers have BOD concentration between 2-4mgO₂/l or greater. This indicates the importance of taking steps to recycle biowaste through enhanced biowaste treatment. This could be accomplished by incorporating bioeconomy principles into water-smart technologies to avoid water pollution of rivers. There is no data available from WP2 on phosphate and nitrates concentrations which could also be used to measure environmental health.

4.7. Romania

4.7.1. Research and Innovation

About 60% of the national renewable energy mix comes from primary solid fuel, followed by approximately 20% from hydropower and 10% from wind. The other renewable energy forms are less than 5% each. There is a low share of renewable energy in transport of 7% which creates room for research and development to increase their contributions. This also depicts the fossil-dependent transport sector and other forms such as electric-powered transport. The share of renewables in electricity is 42% which is above the average of all target countries (31%) but lower than the average (50%) of Austria, Sweden, Croatia, Romania, and Latvia. The share of renewables for heating and cooling accounts for 27% which is below the average of all target countries (30%), Sweden (66%), and Finland (55%). This implies there is room for improvement to increase the share of renewables which could be achieved through research and development (R&D). R&D has helped in technology development in various sectors including agriculture, forestry, food, wood, and bioenergy. R&D will require investment and upscaling of technologies and diversification into other relevant sectors to support the bioeconomy. Implementation challenges include knowledge barriers to research and development.

4.7.2. Society

The bioeconomy is taken into consideration in numerous Romanian national documents. The aforementioned strategies are included: the Integrated National Energy and Climate Plan, the Long-Term Strategy for Reducing Greenhouse Gas Emissions, the Industrial Policy, the National Strategy for Forests 2030, the National Strategy for Circular Economy, and Ordinance No. 6 for the reduction of the impact of plastic products, and the Research Development and Innovation (RDI) Strategy.

These national strategies/plans have helped to implement some initiatives in agriculture (5), forestry (1), food (2), wood (1), and bioenergy (1). To support the bioeconomy sector, these small number of initiatives need to be expanded. Organic farming is an important component of the agricultural sector, as well as 'short agri-food' and 'bioenergy'. Romania took part in the BIOEASTsUP initiative, which resulted in the development of a concept paper aimed at bioeconomy implementation. In addition, seven regions in Romania have published bioeconomy strategies at varied levels. Implementation obstacles include a lack of awareness and policy issues. Hence, there is a pressing need for robust partnerships among stakeholders to formulate a national vision for the national bioeconomy strategy. The Sustainable Development Department in Romania is identified as one of the key stakeholders within the Bioeconomy Hub. It is tasked with maintaining ongoing communication with various ministries, including the Ministry of Agriculture and Rural Development, the Ministry of Environment, Waters, and Forest, the Ministry of Economy, Entrepreneurship, and Tourism, and the Ministry of Research, Innovation, and Digitization. This role ensures their active involvement in the development of the national bioeconomy strategy.

4.7.3. Finance and Investment

Romania has approximately 11,750 euros GDP per capita in 2019. With an average contribution of approximately ($\pm 38\%$), agriculture makes the largest contribution to the bioeconomy sector, surpassing the average of 21% made by all CEE2ACT countries and the Baltic States. About 35% come from food, beverages, and tobacco, which is less than the 41% average contribution from all target countries. The contribution of wood, wood products, and furniture to the bioeconomy sector is approximately 12%. The other equally important sectors contribute very little to the bioeconomy. Investment opportunities are abundant in important industries like forestry, food, wood, bioenergy, and agriculture. The sectors provide an opportunity for industrial symbiosis to sustain the bioeconomy sector. Implementation challenges include financial barriers and a lack of infrastructure and capacities.

4.7.4. Environment

Less than 20% of waste generated is recovered by households showing lower recovery rates and bioeconomy potential in the waste sector. The baseline study shows that the separate collection rate of biowaste is below the European average of about 50%. This portends a high potential for biowaste collection and the need to improve the collection rate to sustain the bioeconomy. Nonetheless, as biowaste increases there is a need to increase treatment infrastructure for processing of large volumes of waste. Agriculture produces around 86% of biomass, while forestry produces 14%. Over 75% of biomass is used for energy, with the remaining 20% used for other purposes. This places Romania among the target countries with the largest biomass consumption for energy. This could contribute to the transition to a fossil-free energy industry, which can aid in mitigating climate change.

Environmental health indicators show that BOD in rivers was 3.48mgO₂/l and phosphate levels 0.08mgPO₄/l. The phosphate level is within allowable limits, but the BOD indicates a high (but not heavy) pollution level. But 70% of rivers have BOD concentrations between 2 and 4 mgO₂/l or higher, and 30% of rivers have BOD ranges less than 1.4 to 2 mgO₂/l. Good environmental practices should be used to improve environmental health as these indicators point to a decline. The high BOD level requires measures for mitigation using bioeconomy principles such as cleaner production processes. Circularity can help reduced the amount of biowaste generated through industrial symbiosis which may end up posing environmental risk. This is the surest way of improving biowaste management and building a prosperous bioeconomy. Data on nitrate concentrations in groundwater (WP2) are not available.

The country has about 38% arable land and about 30% forest land area constituting the primary sectors of the bioeconomy. The country has about 2.5% share of organic farming in the total agricultural land area. Organic farming can help protect the environment against climate change and should be encouraged.

4.8. Serbia

4.8.1. Research and Innovation

The renewable energy compositions are primary solid fuel (about 50%), over 35% hydro, and wind (5%). These form the majority in the national renewable energy mix. Geothermal and biogases are less than 5% each requiring research and innovation in these less developed areas including solar and other renewable energy forms that are inadequate. The baseline study showed a lack of data on Serbia's dependence on renewable energy for transport. Obviously, the major forms of transport depend on non-renewable energy technologies. The study also showed a lack of data on the share of renewables in electricity, heating, and cooling as provided by all target countries and the Baltic states. However, the implementation of some national initiatives depicts that there are available natural biological resources that could be transformed through research and innovation into renewable energy technologies. The initiatives were indicated in agriculture, organic waste, food, wood, pulp and paper, and biotechnology which depends on resources from the bioeconomy. Generally, inadequate data constitutes a major knowledge barrier to RDI and bioeconomy implementation in Serbia.

4.8.2. Society

The concept of circular economy is analyzed and elaborated in four National strategies, namely the Agriculture and Rural Development Strategy 2014-2024, the Sustainable Urban Development Strategy by 2030, the Industrial Policy Strategy 2021 to 2030, the Smart Specialization Strategy 2020 – 2027, and two National programs (Circular Economy Development Program 2022-2024 and Waste Management Program 2022–2031). Some projects have been implemented as a result of these plans and strategies in agriculture (2), organic waste (1), food (1), wood (1), pulp and paper (1), and biotechnology (1). Even though these initiatives are few, they can serve as a good starting point that can lead to expansion and diversification into other sectors to support the national bioeconomy.

The country is poised to implement bioeconomy as evidenced by cooperation with ministries, meetings, and conferences. This collaboration has yielded activities such as the Regional Agency for the Development of Small and Medium Size Enterprises Alma Mons, the Group for Circular and Green Economy of the Ministry of Environmental Protection, and meetings at the Ministry of Agriculture, Forest and Water Management of Serbia. Notwithstanding these efforts, experts in Serbia have stated that there is currently no National Bioeconomy Strategy, but there is a roadmap for a circular economy. However, the challenges to bioeconomy implementation are the unavailability of data on the bioeconomy status in Serbia and on the EU Bioeconomy Country Dashboard. Other challenges include a lack of awareness creation and legal barriers.

4.8.3. Finance and Investment

The GDP per capita of Serbia in 2019 was approximately 6,580 Euros. The contribution of the various sectors to the total bioeconomy turnover is not available as evidenced by the lack of data from the baseline assessment report (WP2) on Serbia. This is one of the challenges to the implementation of bioeconomy as explained in section 4.8.2 (Society).



Despite this, the country has been able to implement few initiatives in agriculture, organic waste, food, wood, pulp and paper, and biotechnology. The few initiatives will require some investment to support the expansion of the bioeconomy sector. Implementation challenges include financial barriers and a lack of infrastructure and capacities.

4.8.4. Environment

There was no data available on biomass production and consumption in Serbia. However, primary solid fuel (50%) is the main form of renewable energy meaning a high level of biomass comes from forestry. There is also a lack of data on the separate collection rate of biowaste as provided by all target countries and that makes it challenging to make a good assessment of the potential for biowaste collection. The lack of data on biowaste generated and recovered also makes it challenging to quantify the available biowaste resources that will sustain the national bioeconomy. The country has about 29% arable land and about 32% forest land area, constituting the primary sectors of the bioeconomy. This shows that bioeconomy resources will come from the agriculture and forestry sectors since these sectors constitute 61% of the land area. About 39% of the land area is other land. The country has no data on the share of organic farming in the total agricultural land area, even though organic farming has less carbon footprints and is environmentally-friendly. Regarding environmental health, there is no available data from the baseline study (WP2) that could be used as indicators to measure the status of environmental health in Serbia.

4.9. Slovakia

4.9.1. Research and Innovation

Primary solid fuel is the main form of renewable energy accounting for over 60%, followed by more than 15% of hydro, and 5% biogases in the national renewable energy mix. The other less developed sectors contribute less than 5% each for pure biodiesel, pure biogasoline, renewable municipal waste, solar thermal and photovoltaics, and geothermal. There is a low share of renewable energy in the transport sector of 7% due to the over-dependence on combustion engines, electricity-powered transport, and the use of synthetic fuels. The share of renewables in electricity is 21% which is below the average of all target countries (31%). The share of renewables for heating and cooling constitutes 10% which is below the average of all target countries (30%). The country has implemented a wide range of initiatives in agriculture, organic waste, food, wood, biobased chemicals, ecosystem services, and bioenergy. This has the potential to increase research, development, and innovation across sectors including renewable energy technologies. Challenges include a lack of implementation.

4.9.2. Society

Many national documents in Slovakia have highlighted aspects of the bioeconomy for consideration. This includes CAP Strategic Plan 2023–2027, Strategy for Forestry Development in Slovakia, Greener Slovakia– The Strategy of the Environmental Policy by 2030, Integrated National Energy and Climate Plan 2021 – 2030, Low-carbon



Strategy by 2030-2050, Strategy for use of Renewable Energy Resources, Economic Policy Strategy by 2030. These strategies and plans have helped to implement a wide range of initiatives in agriculture (4), organic waste (1), food (3), wood (1), biobased chemicals (1), ecosystem services (2) and bioenergy (3). According to Slovak experts, there is no dedicated National Bioeconomy Strategy in Slovakia but there is a roadmap for Circular Bioeconomy and 'carbon agriculture' as a major field in the planned strategy. However, the Ministry of Agriculture and Rural Development of the Slovak Republic is collaborating with ministries, the Slovak Agriculture and Food Chamber, and having meetings, and conferences to support the development of bioeconomy. Implementation challenges of bioeconomy include lack of awareness and inadequate collaboration, policy and legal barriers as well as lack of implementation. Commitment from representatives of the Ministry of Agriculture to finalize the Bioeconomy Roadmap until the end of 2024 is acknowledged. Synergies between various bioeconomy initiatives could serve as inspiration and support for bioeconomy development.

4.9.3. Finance and Investment

The GDP per capita of Slovakia in 2019 was approximately 17,390 Euros (based on calculation from normalized to NL, which has the highest GDP with EUR 47,000 per capita). Food, Beverage, and Tobacco constitute the most significant contribution to the bioeconomy sector at about ($\pm 39\%$), lower than the average contribution of 41% across all CEE2ACT countries, including the Baltic States. Agriculture contributed ($\pm 24\%$), which is higher than the average contributions of all target countries of 21%. Wood, wood products, and furniture constitute about 14% (higher than the average of all target countries of 13%) of the contribution to the bioeconomy sector. Pulp and paper contribute about 11% higher than the average of all target countries. The other equally essential sectors are comparatively low regarding their bioeconomy contributions. Implementation of a wide range of initiatives in different sectors can help draw investment opportunities in agriculture, organic waste, food, wood, biobased chemicals, ecosystem services, and bioenergy. Implementation challenges include financial barriers.

4.9.4. Environment

The country has about 10% share of organic farming in the total agricultural land area. Organic should be encouraged because of its lower impact on the ecosystem and the environment. Indicators of environmental health, BOD in rivers ($2.17\text{mgO}_2/\text{l}$), phosphate in rivers ($0.07\text{mgPO}_4/\text{l}$), nitrate in groundwater ($14.38\text{mgNO}_3/\text{l}$). The BOD shows moderate pollution levels, and the phosphate and nitrates are within acceptable limits. Monitoring of water bodies and cleaner production should be encouraged to maintain good environmental health indicators to safeguard the ecosystem.

About 60% of biomass is obtained from agriculture and 40% from forestry. Over 60% of biomass is consumed for energy (kt of dry matter) and more than 30% for different materials (kt of dry matter). Less than 50% of waste generated is recovered by households. This shows great potential for waste management to support the bioeconomy. The separate collection rate of biowaste is below the European average of about 50% showing a high potential for biowaste collection and the need to improve the



collection rate to sustain the bioeconomy. The country has about 28% arable land and about 40% forest land area constituting the primary sectors of the bioeconomy.

4.10. Slovenia

4.10.1. Research and Innovation

The majority of Slovenia's biomass production is derived from forestry and agriculture. However, the contribution from fisheries remains notably limited despite the country's recognition of the blue economy as an important sector. Primary solid fuel is the main form of renewable energy, with about 50% and over 35% hydro in the renewable energy mix. Biogases, heat pumps, solar thermal and photovoltaics and geothermal renewable energy forms contribute less than 5% each. There is a low share of renewable energy in the transport of 3% due to the high dependence on non-renewable forms of energy such as fossil fuels. Therefore, the limited sources of renewable energy and its uses provide the opportunity for research and development in this area. Already, research and development initiatives are being implemented in Slovenia including the development of biochar potential (Alps4GreenC). The share of renewables in electricity is 32% which is above the average of all target countries (31%) but lower than the average (50%) of Austria, Sweden, Croatia, Romania and Latvia. The share of renewables for heating and cooling constitutes 35% which is above the average of all target countries (30%) but below Sweden (66%) and Finland (55%). Even though Slovenia is performing above averages in renewable energy shares in electricity, heating and cooling, and low shares in transport, there is much potential for increase which could be achieved through research and innovation. The challenge to implementation is mainly knowledge barriers.

4.10.2. Society

There is a greater tendency of bio-based source utilization in electricity (32%), heating and cooling (35%) in the country, indicating the potential support of the society towards green energy.

Slovenia has implemented national strategies in agriculture, forestry, climate/zero-emission energy, circular economy, and blue economy. Some of these include the Strategic Plan of the Common Agricultural Policy 2023-2027 (CAP), National Forestry Strategy, A new approach for a sustainable blue economy, Roadmap towards the Circular Economy in Slovenia and Framework Program for the Transition to a Green Economy. A collaborative inter-ministerial project coordinated by Ministry of Public Administration together with Ministry of environment, climate and energy has been implemented, creating a portfolio of integrated interventions across the five value chains - built environment, food, forest-based & wood, mobility and manufacturing. Ministry of agriculture, Forestry and Food and Ministry of economy tourism and sports will also be engaged, which showed the commitment of the government for finding solution of climate change in multiple sectors. In addition, a lack of a strong connection between ministries and other stakeholders, such as industry actors is recognized as a main obstacle in Slovenia. Other challenges include lack of motivation, policy and legal

barriers. Furthermore, Slovenia's stakeholders recognized the potential drawbacks for bioeconomy principles implementation especially for research and innovation and industrial sector investment, in particularly during this transition phase.

As many as 10 national initiatives have been compiled, of which they are attributed to agriculture (2), organic waste (2), food (2), bio-based chemicals and materials (2) and bioenergy (1).

4.10.3. Finance and Investment

The country has allocated its budget to key areas of bioeconomy-related strategies across various sectors. Notably, the largest contribution of sectors to the total bioeconomy turnover came from the Food, beverage, and tobacco sector, accounting for approximately 30%, lower than the average contribution of 41% across all CEE2ACT countries, including the Baltic States. This was followed by Agriculture and Wood, wood products and furniture, at over 15%, and Bio-based chemicals and materials, which approached 14%. The percentage contribution of agriculture was lower than the average agriculture contributions of all target countries of 21%. On the contrary, the percentage contribution of wood, wood products and furniture was higher than the average contribution of all target countries of 13%. Pulp and paper contributed about 11% higher than the average contributions of all target countries of 9%. The other equally essential sectors are comparatively low regarding their bioeconomy contribution. There is an opportunity to invest and diversify into other sectors with low bioeconomy turnovers. This implies that the country should look for more opportunities in the biowaste/raw material streams to sustain the bioeconomy. The low status of some forms of renewable energy sector is an opportunity to invest in developing those sectors. The implementation challenges include financial barriers due to the lack of funding and investment needed to build infrastructure. Additionally, in 2019, the GDP per capita in Slovenia amounted to 25,650 Euros.

4.10.4. Environment

The forest area constitutes most of Slovenia's land (about 60%), while agriculture occupies about 10% of the total land area. About 45% of biomass is obtained from agriculture and 55% from forestry. Over 55% of biomass is consumed for energy and more than 40% for different materials. The country's share of organic farming production in the EU is about 10% of the total agricultural land area. This is quite low considering the importance of organic farming as climate-smart agriculture for climate change adaptation and mitigation. In Slovenia, the origin of renewable energy is mainly from primary solid biofuels, followed by hydro-source. Only 3% of the renewable energy is used for energy sources for transportation, considerably less than bioelectricity or heating and cooling. More than 50% of biowaste generated by households has been managed and re-utilized, indicating that households do not recover more than 40% of waste generated. This shows great potential for waste management to support the bioeconomy. The separate collection rate of biowaste is about 80% above the European average of about 50%, showing a high potential for biowaste collection. Indicators of environmental health in Slovenia include BOD in rivers (0.86mgO₂/l), phosphate in rivers (0.02mgPO₄/l), and nitrate in groundwater (18.25mgNO₃/l). The BOD of less than 1

mgO₂/l shows that the rivers are generally clean without pollution, and the phosphate and nitrate levels are also within acceptable limits. This shows good environmental health and should be sustained.

5. ANALYSIS RESULTS OF BIOECONOMY STRATEGY VISIONS AND OBJECTIVES

5.1. Visions

Using the described methods in chapter 2, results of the analysis are presented in this chapter. The visions are sorted according to the most frequent repeating topics within the each of our four categories: (1) bioeconomy strategies for research and innovation (2) social challenges, considering gender aspects and re-skilling needs, (3) finance and investment, and (4) environment.

5.1.1. Bioeconomy Strategy for Research and Innovation

Research

The visions for bioeconomy research and innovation predominantly emphasized research as a key element. This aligns with the thematic nature of the visions and is a positive finding, given that research is included in the visions of all surveyed countries. It is mostly linked to the demand for innovation to further expand the bioeconomy sector. It is also indirectly linked to new technological solutions and the process of digitization. In total, these elements are mentioned in the visions of all countries, i.e. Poland, Bulgaria, Greece, Hungary, Romania, Slovakia, Croatia, Czech Republic, Serbia, and Slovenia.

Cooperation

Another prominent aspect of the visions is some form of cooperation. This concerns the following countries: Poland, Bulgaria, Hungary, Romania, Czech Republic, and Slovenia. In most cases, cooperation in the sense of research and innovation represents some kind of collaboration between research bodies and representatives of business and the public sector. It also highlights the requirement that all the actors concerned be allowed to participate in the innovation process.

Economic aspects

A total of six countries, i.e. Poland, Bulgaria, Greece, Croatia, Czech Republic, and Slovenia, also had an economic aspect in their visions in the context of a bio-economic strategy for research and innovation. This usually involves some form of more efficient production processes, increased productivity, increased economic prosperity or the overall development of economies according to concepts such as circular economy or green economy. These concepts have a common goal, i.e. achieving sustainability.

Management of renewable resources

Another important part of the countries' visions was better use of renewables in the context of bio-economic strategies for research and innovation. Mostly this relates in some form to the requirement for innovation to contribute to more efficient

management of renewable resources. There is also a requirement to expand the use of bio-based resources or cascade biomass. Last but not least, the use of bio-waste as an input to production processes was also included. The countries concerned were Poland, Greece, Hungary, the Czech Republic and Serbia.

Education

In one case, i.e. Serbia, the vision described a requirement for a focus on education through new degree programs that would focus on the bioeconomy in their content.

Table 2 depicts the compilation of the visions in research and innovations of the participating countries.

Table 2. Vision 5.1.1 Research and Innovations

Country	Research	Cooperation	Economic aspects	Management of renewable resources	Education
Bulgaria	x	x	x		
Croatia	x		x		
Czech Republic	x	x	x	x	
Greece	x		x	x	
Hungary	x	x		x	
Poland	x	x		x	
Romania	x	x			
Serbia				x	x
Slovakia	x				
Slovenia	x	x	x		

5.1.2. Social challenges, considering gender aspects and re-skilling needs

Equity

In almost all countries surveyed, an equity element on multiple levels appeared in the visions of social challenges in the context of the bioeconomy. It usually represents some form of equal opportunity for all segments of society. Furthermore, equality in terms of gender or age is often mentioned. The bioeconomy as a sector is supposed to be inclusive and involve the whole of society in this respect. In particular, equal conditions are to be reflected in education and employment opportunities. This applies to the following countries: Poland, Bulgaria, Greece, Hungary, Romania, Slovakia, Croatia, the Czech Republic and Serbia.

Education

Another important topic in the visions was education. In total, this concerned the following countries: Greece, Hungary, Romania, Croatia, Serbia, and Slovenia. Education includes general awareness of the bioeconomy. This one is most often

associated with equal access to education, but also with the requirement to cultivate the future generation as leaders and pioneers in the bioeconomy. Overall, there is a strong emphasis on knowledge capital, which is linked to the development of the bioeconomy sector.

Economic aspects

Poland, Bulgaria, Hungary, Romania, Slovakia and the Czech Republic also had in their visions an economic aspect in the context of social challenges. In most cases, addressing social challenges is a precondition for economic prosperity and sustainable economic growth. This fact is based on the very definition of sustainable development. According to the countries surveyed, an inclusive society and a level playing field can be seen as a condition for the achievement of economic goals.

Employment

Another important feature was employment in the context of social challenges. In total, this element appeared in the following countries: Greece, Hungary, Croatia, Czech Republic, and Serbia. In summary, based on the visions, it can be concluded that job creation in the bioeconomy is an important social challenge that can significantly help in the fight against unemployment given the untapped potential that the bioeconomy opens up.

Cooperation

In two cases, the request for cooperation was also mentioned. These concerned Hungary and Slovenia. This referred to cross-sectoral dialogue as an important flow of information and knowledge, but also to the requirement for synergy between the disciplines and stakeholders involved.

Summary of visions of the social changes is presented in the following Table 3.

Table 3. Vision 5.1.2 Social challenges

Country	Equity	Education	Economic aspects	Employment	Cooperation
Bulgaria	x				
Croatia	x		x	x	
Czech Republic	x	x		x	
Greece	x	x	x	x	
Hungary	x		x	x	x
Poland	x	x			
Romania	x	x	x		
Serbia		x		x	
Slovakia	x		x		
Slovenia		x	x		x

5.1.3. Finance and investment

Funding

In the finance and investment category, all countries, i.e. Poland, Bulgaria, Greece, Hungary, Romania, Slovakia, Croatia, Czech Republic, Serbia, and Slovenia, had an element of funding in their visions. Overall, different forms and levels of funding are included here. Both national and regional funds, as well as European ones, are mentioned here. Within these financial instruments and sources, the visions include both private and public forms of financing.

Cooperation

Cooperation, which is mentioned in a large part of the vision, is significantly related to the previous one. Cooperation in the framework of financing and investment mainly represents a combination of different sources and cooperation in the framework of financing, which will make finance more accessible for the development of the bioeconomy. This applies to the following countries: Poland, Bulgaria, Hungary, Slovakia, Croatia, Czech Republic, Serbia, and Slovenia.

Economic aspects

In one case, in Greece, the vision described a future expected state that can be achieved through sustainable financing and investment - economic growth in the context of a green and circular economy.

Table 4 shows the key insights in finance and investment of the target countries

Table 4. Vision 5.1.3 Finance and investment

Country	Funding	Cooperation	Economic aspects
Bulgaria	x	x	
Croatia		x	
Czech Republic		x	
Greece	x		x
Hungary	x	x	
Poland	x	x	
Romania	x		
Serbia			
Slovakia		x	
Slovenia		x	

5.1.4. Environment

Management of renewable resources

In the category related to the environment, better use of renewable resources was a very frequent part of the states' visions. In particular, a demand was formulated here

for the sustainable use of renewable resources for production processes, which is gradually moving towards an angle-neutral economy. Specific types of natural capital, such as forest biomass, which must be treated sustainably and sparingly, were also listed here. It concerned the following countries: Poland, Bulgaria, Greece, Hungary, Romania, and Serbia.

Nature conservation

Nature conservation is also related to the previous category. It was included in the visions of the following countries: Poland, Greece, Romania, Czech Republic, Serbia, and Slovenia. This requirement was mainly formulated in the context of the definition of sustainable development concerning the preservation of natural capital for future generations as much as possible in an unchanged form. In their visions, countries are aware of the need to preserve the environment in an unchanged form, to which the expansion of the bioeconomy should contribute.

Economic aspects

The economic aspects were found in the visions of the following countries: Poland, Bulgaria, Hungary, Slovakia, Croatia, Czech Republic. Concerning the definition of sustainable development, this aspect has been integrated into the context of the environmental pillar. Economies that will emphatically protect the environment are connected with the concepts of circular economy and bioeconomy in particular, which represent an environmentally friendly form of economic growth and development.

Research

In the case of Slovenia, the role of research was also included in the vision. Research and innovation here play a significant role in reducing the negative impact on the environment.

Education

In the case of Bulgaria, the vision contained a requirement for high ecological awareness across society as a means of developing the economy in line with the preservation of a healthy environment.

Visions of the environment sector drafted by the participating countries is shown in Table 5.

Table 5. Vision 5.1.4 Environment

Country	Management renewable resources	Nature conservation	Economic aspects	Research	Education
Bulgaria	x		x		x
Croatia			x		
Czech Republic		x	x		
Greece	x	x			
Hungary	x		x		
Poland	x	x	x		
Romania	x	x			
Serbia	x	x			
Slovakia			x		
Slovenia			x	x	

5.2. Objectives of the bioeconomy strategy

Similar to the visions, objectives are classified according to the most frequent repeating topics within the each of four categories, as follows: (1) bioeconomy strategies for research and innovation, (2) social challenges, considering gender aspects and re-skilling needs, (3) finance and investment, (4) Environment.

5.2.1. Bioeconomy strategies for research and innovation

Cooperation

When speaking about objectives of bioeconomy strategies in the field of research and innovation the most frequently cited category is “cooperation”. This category is mentioned by a total of 7 partner countries - Croatia, Czech Republic, Greece, Poland, Romania, Slovenia and Serbia.

The inter-sectoral cooperation seems to be the most important one from the point of view of Poland, Romania, Serbia and Slovenia. Inter-sectoral cooperation links together the government, the private sector, educational and research organizations and local communities and it is related both to traditional as well as new branches. At the same time, it is important to cover all types of key stakeholders.

The second most frequent type of cooperation mentioned is the strengthening and development of relations between research and development/R&D (science, academic sector) on the one hand and the manufacturing sector on the other one - it is mentioned in the case of Croatia, Czech Republic, Romania and Slovenia. This is sometimes simplistically referred to as cooperation between the public and private sectors. This answer reflects the rather complex situation concerning the applicability and speed of the transfer of science and research results to production practice. In the case of the Czech Republic and Slovenia is mentioned also the active role of regions and

the need to respect their specific conditions.

The international cooperation and knowledge sharing using the advantage of digitalization which is linking the countries as so as EU as an umbrella institution but also different national institutes and organizations involved in bioeconomy development is mentioned by partners from Greece and Slovenia.

Poland and Slovenia recommend focusing on cooperation in education, both in terms of public awareness and school education of the new opportunities in different sectors offered by bioeconomy and its expected positive impacts on global change.

Slovenia proposes to establish strong key companies in each bioeconomy sector to ensure biomass supply while maintaining the influence of the national government on the development of the circular bioeconomy.

Poland suggests increased cooperation and development of evidence-based policies including research and innovation aspects.

Research

The category related to conditions and research topics is important from the perspective of partners from Bulgaria, Croatia, Czech Republic, Greece, Hungary, Poland, Serbia, Slovakia and Slovenia.

A general focus on new trends and the use of available data in research in the field of bio-based technologies, and biorefineries with the support of international cooperation appears in the objectives stated by Bulgaria, Hungary, Poland, Serbia and Greece.

Identification, mapping and focus of future research to national potential for bioeconomy (including waste) and key players in this field is a priority for the Czech Republic, Hungary and Serbia. Bulgaria, Croatia, and Slovakia point to the need for general support of bioeconomy in the field of research and innovations and the growth of the importance of this topic within the academic sector.

For Greece and Slovenia, it is very important to find and promote good practices in circular economy models, encouraging the reuse, recycling, and upcycling of bio-based products and waste materials.

Education

Education as part of bioeconomy R&D objectives plays a key role in the opinions of partners from Croatia, Greece, Poland, Romania and Serbia.

Education and training for key stakeholders operating in the field of bioeconomy is preferred by Poland, Serbia, Croatia, and Greece.

The increasing awareness in sense of understanding bioeconomy and circularity principles for the general public via school education and good practice promotion is proposed especially by Poland and Romania.

Economic aspects

In particular, Bulgaria, Poland, Romania and Slovenia mention economic goals in their draft bioeconomy strategy for science and research.

Bulgaria has probably the most elaborate proposal in this respect, including modernization and greening of facilities, international cooperation, and retraining of the workforce leading to an overall strengthening of its position on the international market.

Strengthening the competitiveness of the bioeconomy (or its products) is an important



objective for Bulgaria and Romania.

Slovenia suggests an orientation towards the promotion of biorefineries in agri-food SME enterprises. Poland proposes to increase agricultural productivity through sustainable intensification.

Policy

The improvement of policy linked with future circular bioeconomy development as a part of R&D objectives is important from the point of view of the Czech Republic, Hungary, Serbia and Slovenia.

The prevailing opinion represented by Serbia and Slovenia shows the need of a national bioeconomy strategy and related legal measures. The Czech Republic suggest identifying legislative barriers followed by correction of the legal framework. For Hungary is important to establish the dialogue among the key stakeholders.

Funding

The issue of funding for science and research in the field of circular bioeconomy is part of the objectives proposed by the Polish, Greek, Slovak and Serbian partners. In essence, all of them propose to ensure stable funding for research and innovation in the bioeconomy and related entrepreneurship, as well as education.

Management of renewable resources

Optimization of the use of biomass is the key topic for the Czech Republic, Hungary, Poland and Slovenia.

Three countries - Hungary, Poland and Slovenia emphasize the importance of cascade use of biomass. The Czech Republic suggest using yet untapped biomass (i.e. without adequate technical infrastructure) resources strictly within the framework of the circular bioeconomy.

Table 6 depicts the summary of objectives in research of innovations of the participating countries

Table 6. Objective 5.2.1 Research and Innovations

Country	Cooperation	Research	Education	Economic aspects	Policy	Funding	Management of renewable resource
Bulgaria		x					
Croatia	x	x	x	x			
Czech Republic	x	x			x		x
Greece	x	x	x	x		x	
Hungary		x			x		x
Poland	x	x	x	x		x	x
Romania	x		x	x			
Serbia	x	x	x	x	x	x	
Slovakia		x	x			x	
Slovenia	x	x			x		x

5.2.2. Social challenges, considering gender aspects and re-skilling needs

Education

Undoubtedly the main challenge among the social challenges proposed to be involved in bioeconomy strategies is education – this is the common opinion of the target countries.

The biggest share of these activities is represented by increasing of social awareness – sometimes in its very general form, sometimes aimed especially on specific topic (ie. biomass cascading, biogas, biomethane, global change, human diet, mutual substitutability of products) or on special group (usually youth). Routine education of bioeconomy as a study field or course in various types of schools, including universities, is the second most frequent subgroup of the education category - according to the view presented by Croatia, Hungary, Romania, Slovakia, Slovenia and Serbia. It includes both education of young generations and, for example, lifelong learning. The importance of bioeconomy education for future employment is sometimes emphasized.

Typically, the orientation of education is more towards a general view of the bioeconomy and its potential, but there are also demands for specific types of education within (organic) agriculture, bio-based products, forestry or better use of waste.

In the opinion of experts from Romania, Slovenia, Slovenia and Serbia, the training of the workforce and entrepreneurs in the bioeconomy is a very important aspect that will guarantee a sufficient number of qualified employees and entrepreneurs in this sector. Again, this is mostly a more general form of information about the bioeconomy and its potential, in some cases focused on specific sectors (forestry) or issues (climate change). The training and education of specialists essentially build on the previous section, according to the opinion represented by the Czech Republic, Hungary and Slovenia - it more or less states the future need for a highly skilled workforce and assumes that it can be the key to solving some of the problems e.g. employment in rural areas.

The rest of the opinions is related to single topics such as the redefinition of certain wastes as usable biomass which should be as a topic publicly discussed to be widely accepted Czech Republic.

Education programmes for the development of digital skills to be used in agriculture was mentioned by Hungary.

Introducing changes in the socio-economic model to enable the transition to a sustainable circular bioeconomy presented Poland.

Cooperation

The second most frequented subcategory within education is cooperation its importance is well known, especially in Bulgaria, Croatia, Greece, Poland, Serbia and Slovenia. Very often is mentioned the need for cooperation among institutions or representatives of different sectors - such as (higher) education, business and research (science) – this is the case in Bulgaria, Greece and Poland. The emphasis is also taken on cooperation with local communities with respect to their specific features and needs (Croatia, Greece). A little bit different area is the cooperation that should ensure an adequate bioeconomy policy - identify potential stakeholders (Serbia), ensure cross-

ministerial cooperation (Slovenia) or bring bioeconomy to such an important sector as agriculture (Serbia).

Equity

Equal opportunities within bioeconomy as an important aspect especially in employment, education and decision-making and overall supportive environment creation from the gender, age and vulnerable communities' point of view are the crucial points for Croatia, Greece, Poland, Serbia, and Slovenia. The gender equality is mentioned by the partners from Croatia, Greece, Poland and Serbia. The issue of young generation - its involvement into bioeconomy as possible solution of the young people's migration from rural areas is a part of the opinion of Poland and Slovenia. The equal opportunities for involvement of vulnerable communities into bioeconomy are mentioned by Greece.

Employment

Job creation and reskilling opportunities are important features of bioeconomy sector from the point of view of Croatia, Czech Republic, Poland and Romania. There are different issues mentioned that probably could be solved via the job creation in bioeconomy solve such as employment generally or employment in rural communities, jobs for the young generation or solutions for work-life balance. There is also mentioned the possible use of economic instruments to create and save job in bioeconomy sector (Croatia, Romania, Czech Republic). Re-skilling of employees for work in bioeconomy sector as a tool for the prevention of unemployment is mentioned by Croatia and Poland.

Policy

Czech Republic, Greece, Hungary and Slovenia suggest creating new and/or changing several existing policy measures related to social changes as a part of bioeconomy strategies which were not yet mentioned in the above topics. The main tasks in this field seem to be respecting food security (which is also part of SDGs) – mentioned by the Czech Republic and Hungary - in case of Hungary there also the support of local organic farming added. Support of work-life balance, flexible family-friendly work arrangement is proposed by Greece. Identification and rectification of political bottlenecks as well as simplification of legal and permit procedures are topics suggested by Slovenia.

Economic aspects

Other suggested parts of social changes in national bioeconomy strategies rely on a general economic view. In the opinion of Czech partners, the circular bioeconomy offers (at least partially) the possibility of maintaining living standards while reducing the burden on the environment. For Romanian partners is important to enable capacity building along the entire value chains.

The objectives stated by the target countries can be seen in Table 7.

Table 7. Objective 5.2.2 Social challenges

Country	Education	Cooperation	Equity	Employment	Policy	Economic Aspects
Bulgaria	x	x				
Croatia	x	x	x	x		
Czech Republic	x			x	x	x
Greece	x	x	x		x	
Hungary	x				x	
Poland	x	x	x	x		
Romania	x			x		x
Serbia	x	x	x			
Slovakia	x					
Slovenia	x	x	x		x	

5.2.3. Finance and Investment

Policy

As regards objectives of bioeconomy strategies in the field of finance and investment there is the most frequently cited category of policy, which is mentioned by these countries: Croatia, Hungary, Czech Republic, Greece, Serbia, Poland, Romania, Slovenia, Slovakia.

The category of policy could be divided into three additional subcategories: what to support and improve, finding sources of financing and what are the predisposition of successful bioeconomy funding.

Within the first subcategory what to support and improve we could find these subjects: Research innovation, project financing, infrastructure, small-scale biorefineries, and technological modernization mentioned by Poland, Hungary Czech Republic, Greece. Other subjects for financial support are farmers and foresters who work directly with biomass expressed by Slovakia. Another kind of support ought to be new technologies, and infrastructure modernization. Within policy an important topic is also mentioned promoting education in the bioeconomy and its financing. Education is also an important predisposition for the whole bioeconomy acceptance was mentioned by Romania.

Within the subcategory sources of financing, there are mentioned the mixture of the different options in terms of state, private sector, own and international resources. Interestingly, there was a link between cooperation and financing, which means cooperation between these different forms of finance, the combination and diversity of funds in terms of the type of ownership, state or private support, or national and international - Serbia, Croatia. An additional important topic was private financial

support for bio-economic projects and encouraging new forms of investment - Croatia, Poland, Romania. Support innovative companies that take risks - Greece. A financial source could also be the increase of the export of bio-economic products, raising money to finance national bio-economic activities - Serbia.

As the predisposition of successful bioeconomy funding, first of all, strategic policy documents, including bioeconomy strategy itself are mentioned - Croatia, Poland. Another important part is defining priorities for supported projects related to bioeconomy - Hungary. Creation and legislation of bioeconomy standards for new products their legal anchoring and support are also indispensable - Czech Republic. In general, could be said that the existence of a political and economic environment that creates good conditions for the introduction of the bioeconomy, are crucial. Stable economic and political conditions with clearly defined legal boundaries for new investors and forms of business are substantial preconditions for bioeconomy development - Greece, Hungary. These steps should be completed with the use of specific economic instruments, such as tax rebates for new entrepreneurs and better conditions for access to finance, especially for start-up businesses and the involvement of the banking sector, as they are essential to ensure effective implementation. Related to this is also clarification of the situation in "green finance support". It means being more transparent for donors and acceptors, highlighting the profitability and ROI of new technologies in the bioeconomy sector - Slovenia, Hungary, Greece. Not insignificant is also revamp of funding: overhaul of the national public funding structure to be more responsive to current challenges -Hungary, Slovenia, Romania. In one case financial education - awareness raising about financing mechanisms -as an important background for a new type of investments was mentioned - Romania.

Cooperation

The second place as regards the frequency of the mentioned category within the field of finance and investment objectives of bioeconomy strategies is a topic of cooperation. This category is mentioned by eight of the following partner countries: Slovenia, Romania, Greece, Hungary Czech Republic, Bulgaria, Slovakia and Poland. In the case of cooperation, three topics there were stressed: key networking stakeholders, forms of cooperation and main outcomes- improvement gained via cooperation.

Within the subcategory key networking stakeholders were mentioned cooperation among the government, private sector, and research institutions additional collaborative subjects must be financial institutions and regulatory bodies - Bulgaria, Greece.

Cooperation could serve to foster public-private partnerships. An important role plays also a large-scale business supporting private bioeconomy infrastructure and launching pilot biorefineries. In general cooperation between the private and public sector including banks as important stakeholders are the most important - Bulgaria, Romania, Greece, Slovenia, Slovakia.

Cooperation could have different forms, the most recommended are: Creating synergies and complementarities between funds, integrating projects with private and public stakeholders - Romania create open-access pilot Infrastructure and redirecting focus from trivial debates to initiate bold and impactful steps that will elevate the circular bioeconomy - **Slovenia**. Building and leveraging the first of its kind bioeconomy

entrepreneurship ecosystem – Poland. In general, we could conclude that financial support for all forms of cooperation and networking among key stakeholders is essential, at least at the beginning - Czech Republic.

In the subcategory of the main outcomes from a different form of cooperation, we could find: Maintaining a robust logistics and supply chain infrastructure for the seamless movement of bio-based products - Greece, Bulgaria. Leverage shared resources and increase the accountability knowledge, and funding for bioeconomy projects (Greece), establish clear standards for green finance, ensure transparency and accountability in bioeconomy-related investments and promote a pipeline of investable projects in collaboration (Hungary, Poland). Invite investors and banks to a discussion on how to promote a pipeline of investable projects in collaboration - Slovenia. Foster public-private partnerships to finance and scale bioeconomy ventures, particularly in promoting sectors like bioenergy, sustainable agriculture, and forestry - Hungary.

Funding

The third place as regards the frequency of the mentioned category within the field of finance and investment objectives of bioeconomy strategies is the topic of funding. This category is mentioned by the following partner countries: Bulgaria, Poland, Romania, Czech Republic and Slovenia. The main areas of funding are recommended: research and development - Poland). Funding to support and improve infrastructure for biorefineries and waste management and high technology and innovation - Bulgaria, Poland. Support of small-scale business cases individual projects plus start-ups system to grow bioeconomy sectors - Slovakia, Romania.

A possible source of funding for the circular bioeconomy could be the transfer of funds from subsidy titles that only address part of the issue. Funding must keep in mind that the bioeconomy offers not just economic progress, but also a comprehensive environmental solution - Czech Republic.

Education

The topic of education in relation to the funding of bioeconomy was mentioned just by two countries: Czech Republic and Slovenia.

Financial support for formal and non-formal education in the field of circular bioeconomy - Czech Republic.

Enhance Investment Knowledge: Promote awareness and training on modern investment collaboration models like Joint Ventures. Provide Case studies on investments for deploying circular bioeconomy value chains with business cases Slovenia.

Table 8 shows the target countries' objectives in finance and investment.

Table 8. Objective 5.2.3 Finance and Investment

Country	Policy	Cooperation	Funding	Education
Bulgaria		x	x	
Croatia	x			
Czech Republic	x	x	x	x
Greece	x	x		
Hungary	x	x		
Poland	x	x	x	
Romania	x	x	x	
Serbia	x			
Slovakia	x	x		
Slovenia	x	x	x	x

5.2.4. Environmental dimension in bioeconomy strategies

Management of renewable resources

Within the objectives of bio-economic strategies, the last category examined is the Environment. In the context of the environment, better use of renewable resources was the basis of the objectives for the largest number of countries.

The largest number of countries, i.e. Romania, Serbia, Slovakia, Czech Republic, Hungary, Greece, Bulgaria, and Poland further specify and develop this fact. In particular, it is about circular and sustainable management of natural resources. In this context, the resource is meant to be biomass, wood and other renewable resources. It also mentions, for example, the cascading use of biomass. All of these then lead to general environmental objectives - in particular, minimising the environmental impact and dealing with the negative externalities of economies. The adjectives circular and sustainable correspond to the current discourse within strategic documents at national and EU levels.

The other way to make better use of natural resources was a significant change in waste management. Here a general goal was set, and the targets included a total reduction in waste. This is mainly linked to efforts to produce more efficiently economically, but also ultimately to significantly reduce the environmental impact. This is reflected in the targets of the following countries: Slovenia, Slovakia, Serbia, Bulgaria, and Hungary.

In one case, Slovenia, there was a call for greater energy self-sufficiency and energy innovation to make the energy sector more efficient. In one case, Croatia, a call for more awareness and support for bio-based and nature-based solutions in bioeconomy projects was also described.

Policy

The policies were also subject to many objectives. In total, they appeared in the goals of the following countries: Czech Republic, Bulgaria, Greece, Croatia, Poland, Hungary, and Serbia.

In the policy context, the most important part of this objective is the circular, green and sustainable strategic orientation of economies, leading to a transformation towards a more innovative and competitive form of economy. This one can be found in the following countries: Czech Republic, Bulgaria, Greece, Croatia, and Poland.

Another important part of the policy objectives was waste and better use of resources. These were mainly the following countries: Bulgaria, Hungary, Poland, Greece, and Serbia. In particular, the conclusions of the circular economy concept are confirmed here. More specifically, the policies were described in the case of Poland. This was an energy transformation.

Furthermore, in the case of Croatia and Poland, the focus was also on projects and research.

Global change

Global change was also often included in the goals of countries. These were mainly the following countries: Slovakia, Croatia, Serbia, Greece, Hungary, Slovenia, Poland, Czech Republic.

Global change was most often mentioned in the context of awareness of the bioeconomy and its application, which will bring new resources to solve climate change. This applies to the following countries: Croatia, Greece, Hungary, Serbia, Slovenia, Poland, Czech Republic.

Overall, it can be described that the appeal to solve or mitigate climate change is highly prioritized here. In the case of Slovakia, Serbia, and Poland, the conceptual prerequisites for the application of the circular economy as the main concept of economic development are also significantly included in the goals.

In one case, i.e. in Slovakia, the global change in the goal is contained in the form of innovations and technologies, which, together with the effective use of resources, can achieve the goals of sustainable development.

Economic aspects

The economic aspect of the environment appeared in the goals of the following countries: Romania, Bulgaria, Serbia, Poland, Hungary, and Greece. Most countries, i.e. Serbia, Greece, Bulgaria, Romania, Poland and Bulgaria, had economic elements in the set goals described through innovations or changes towards the circular economy, bioeconomy and sustainability. Changes on the side of companies are included here - more efficient waste management or more gentle use of resources but also changes on the side of consumers. In addition, innovations in the field of energy are also included.

For Serbia, Greece and Hungary, the role of rural areas and agriculture as a location and sector for potential development was then significantly highlighted.

Education

Education in the context of the environment appeared in the objectives of three countries: Croatia, Greece, and Serbia.

In all cases, it was some form of promoting the growth of awareness of the bioeconomy or circular economy. The aim of this awareness-raising is in most cases to try to gain public support for these environmentally friendly solutions, but the amount of knowledge among the professional public and professionals entering the labor market also plays a significant role.

Nature conservation

Natural conservation is found in the states' goals in two ways. First, it is about reducing negative externalities to the environment. Here it is mainly about waste and overall pollution.

The second direction is precautionary measures that protect ecosystems and natural capital. These elements are found in the objectives of the following countries: Serbia, Czech Republic, and Hungary.

Management of renewable resources is of course an important part of the environment focus area, where it ranks first in terms of frequency of occurrence. In the Resources and innovation area, on the other hand, it is last, which by no means that research should not address the use of renewable resources in the case of the bioeconomy.

Funding issues are found in the two areas of research and innovation, where it is one of the most important, and finance and investment, where it is of medium importance overall.

Other topics are no longer cross-cutting and are only mentioned in individual areas - Research is seen as the most important group of topics under research & innovation. Equity and employment are medium importance topics under social challenges. Global change (moderately important) and nature conservation (less important) are listed under the environmental themes of bioeconomic strategies.

It can be said that based on our results, policy and education, cooperation and economic aspects are considered the most important in terms of bioeconomic objectives, and are present in at least three of the selected thematic areas, cooperation is completely represented in all four areas.

Objectives of the environment mentioned by target countries are depicted in Table 9.

Table 9. Objective 5.2.4 Environment

Country	Management of renewable resources	Policy	Global change	Economic aspects	Education	Nature conservation
Bulgaria	x	x		x		
Croatia		x	x		x	
Czech Republic	x	x	x			x
Greece	x	x	x	x	x	
Hungary	x	x	x	x		x
Poland	x	x	x	x		
Romania	x			x		
Serbia	x	x	x	x	x	x
Slovakia	x		x			
Slovenia			x			

6. DISCUSSIONS

In the aftermath of the analysis, key themes were identified and prioritized within the categories, and cross-cutting themes were acknowledged for their potential unifying role across multiple categories. Nevertheless, the chosen method has limitations arising from the generalization of the obtained data. Some opinions might be marginalized and could be absent in the subsequent interpretations. Another limitation could stem from the varied forms in which the data were collected. Project partners had the discretion to choose the format for stating visions and objectives, resulting in non-uniform data content with substantial variations.

Finally, based on the collected information, steps in development of the bioeconomy strategy are analyzed and proposed for each CEE2ACT target country.

6.1. Visions of the bioeconomy strategy

The only truly cross-cutting topics for all the defined areas of visions included in bioeconomy strategies (i.e. research & innovation; social challenges; finance & investment; education) are *economic topics*. The importance of economics is in average - medium positions in all four areas. *Cooperation* is a very important element in the three areas, especially for research & innovation and finance & investment, but from the perspective of social challenges, these topics seem not to be so important. *Education* is a common theme in the areas of social challenges, research & innovation and the environment, its importance is evident, especially in the area of social challenges. The issue of *research* plays an important role especially in the area of research & innovation and also in environment area, but is less important in the second case. *Management of renewable resources* is quite very similar situation - it is very important among environmental topics and less important for research and innovation. *Equity* issues seem to be very significant "only" for the area of social challenges, similar is the case of *funding* for the area of finance & investment. Topics linked with *employment* play a role in the case of social challenges. Table 10 depicts summary of cross cutting topics in the visions, categorized into the four predefined sections: research and innovation, social, finance and investment, as well as environment. Different colors used in the table signifies significant areas stated by member states, which are identified by a high frequency of mentions in the visions of the Bioeconomy Strategy.

Table 10. Visions' summary – cross cutting topics

1. Bioeconomy strategies for research and innovation	2. Social challenges, considering gender aspects and re-skilling needs	3. Finance and investment	4. Environment
Research	Equity	Funding	Management of renewable resources
Cooperation	Education	Cooperation	Nature conservation
Economic	Economic	Economic	Economic
Management of renewable resources	Employment		Research
Education	Cooperation		Education

6.2. Objectives of the bioeconomy strategy

Cross cutting topics are also appeared in the objectives of Bioeconomy Strategy, as depicted in Table 11. Similar to the visions, objectives of the Strategy is also classified into the four predefined sections. Likewise different colors in the table represent important areas mentioned frequently in the objectives of the Bioeconomy Strategy.

Table 11. Objectives' summary – cross cutting topics

1. Bioeconomy strategies for research and innovation	2. Social challenges, considering gender aspects and re-skilling needs	3. Finance and investment	4. Environment
Cooperation	Education	Policy	Management of renewable resources
Research	Cooperation	Cooperation	Policy
Education	Equity	Funding	Global change
Economic	Employment	Education	Economic
Policy	Policy		Education
Funding	Economic		Nature conservation
Management of renewable resources			

The subcategory *policy* is mentioned as an objective in all areas, only in each of them it occupies a different importance, as we can see from the order in which it is mentioned in the different areas. The policy seems to be the most important for finance and investment, where it is mentioned right at the top. The prioritization of financial support, its forms, and the definition of the actual sources of funding are issues linked to policy decisions that are expressed in some form of binding documents in the form of plans or strategies. Policy is also important in the environment, where policy comes second. In this case, it is the policy framework for economic activity and innovation that is important in terms of protecting and prioritizing the environment in a bio-economic strategy. Less importance is given to policy in the areas of research and innovation and social challenges, where it is ranked in fifth place.

The subcategory *education* is mentioned in all sections. The importance of education is, again for obvious reasons, very often mentioned in the objectives relating to "research and innovation", where it is in third place. In the area of finance and investment, it is ranked fourth, and in the case of the objectives relating to the environment and the bioeconomy, education is somewhat undervalued and is only ranked fifth.

The sub-category *cooperation* is mentioned as an objective in all areas except the environment. The frequency with which this topic is mentioned varies, as can be seen from the order in which it is mentioned in each area, with research and innovation coming first, social challenges considering gender aspects and re-skilling needs and finance and investment coming second. Cooperation is therefore not only a cross-cutting theme but also a very important theme in terms of the four categories identified. For the three areas of the bio-economic objectives, the topic of *economic aspects* is mentioned. The only area missing is "Finance and investment". However, this is only an apparent omission, as this area is itself economic by nature. In the area of research and

innovation, in line with the environment, economic aspects are listed fourth and social challenges fifth.

The sub-categories *management of renewable resources* are only found in two of the objective areas of the future.

7. CONCLUSIONS

If we compare the groups of topics between the submitted visions of bioeconomy strategies and objectives, we find several common topics that can be seen as stepping stones for a better implementation of these strategies. These common topics are cooperation, education, and economic aspects.

Cooperation is fundamental in terms of the three chosen pillars of bioeconomy strategies - research & innovation: social challenges and finance & investment. Collaboration is understood here very broadly as international, inter-ministerial; inter-company; public-private; science-education-business or fundraising collaboration etc. For the same three pillars (research & innovation; social challenges, finance & investment), the *education* sector is also important - meaning both formal education - general and specialized, and informal education in the form of public awareness.

Economic aspects represent identification with the economic concept, and with a certain degree of simplification can help to understand the role of bioeconomy in society. The central idea is that finance invested in the development of the bioeconomy should return multiple returns in the form of its potential income.

The next set of themes is the *management of renewable resources*, which appears in two pillars - research and innovation and environment, representing the functional fulfilment of the economic and environmental objectives of sustainable development.



8. ANNEXES (Country visions and goals)

Annex 1: Visions and objectives of the target countries

1. Bulgaria

1. Bioeconomic strategies for scientific research and innovation (bioeconomic strategies for scientific research and innovation) In Bulgaria, we have developed the Program "Scientific research, innovation and digitalization for intelligent transformation"		2. Social challenges, taking into account gender aspects and retraining needs (social challenges, taking into account gender aspects and retraining needs). Bulgaria has: A strategy for reducing poverty and especially for social inclusion with a horizon of 2030. NATIONAL STRATEGY FOR PROMOTING EQUALITY OF WOMEN AND MEN 2021-2030		3. finance and investment (finance and investment)		4. environment (environment. In Bulgarian imama: Strategy for transition to a circular economy, 2022 - 2027	
vision	objective(s)	vision	objective(s)	vision	objective(s)	vision	objective(s)
BG 1 Vision - Turning Bulgaria into an innovative, smart, green, digital and knowledgeable country through a new common policy for interaction between scientific research, innovation and technology, as well as increasing international and cross-sectoral cooperation and intensive use of data for accelerated specialization in products and services with high technological and scientific intensity and significant economic impacts for sustainable competitiveness, technological transformation of the economy, increasing resource efficiency and digitization.	BG 1.1 Goal/s Strategic goals: 1. To develop and position Bulgaria as a center of medium- and high-tech innovations in the strategic areas in which the country has established capacity and market positions, as well as to recognize the competence to compete on the world market, increasing the country's national and regional innovation performance; 2. To support the deployment and establishment of a sustainable, modern, dynamic, inclusive, data-driven and globally connected research, innovation and entrepreneurship ecosystem in Bulgaria.	BG 2 Vision - By 2030, Bulgaria is a country in which social inequalities and poverty are limited, created with prerequisites and conditions for accepting and sustainable growth and an opportunity to reduce the quality of life of vulnerable groups. Equality of women and men means equal rights and obligations, equal opportunities for realization and overcoming obstacles in all areas of public life. Women and men should be free to develop their personal capacities and make choices without the constraints of gender stereotypes and discrimination. Ensuring equality between women and men is a fundamental	BG 2.1 Objective(s) - The limited and inefficient cooperation with research organizations, higher education institutions and business, leading to low levels of technology and knowledge transfer and for the commercialization of the results of R&D and innovation in industry and – accordingly – in a slow pace between the production of products with high added value and implementing solutions to address social and economic challenges. The strategic goal of the National Strategy for Promoting the Equality of Women and Men for 2021-2030 is to contribute to the achievement of de facto equality of women and men in Bulgaria,	BG 3 Vision: The development and growth of bioeconomy companies and enterprises, as well as increasing their competitiveness, are of primary importance for the country, which is why serious financial resources and investments are needed. The provision of financial means and investments must be from the state, as well as from private investors, European investment funds and programs, etc. Financing and investing in the development of the bioeconomy will ensure the necessary food security of the country, the fuller use of bioresources and the utilization of biowaste, as well as additional jobs.	BG 3.1 Objective(s) - 1. Invest in infrastructure development, including research facilities, biorefineries and waste management systems, to grow bioeconomy sectors. 2. Cooperation with private investors to build and maintain a stable logistics and supply chain infrastructure for smooth movement of bio-based products. 3. Financing in the development and use of high-tech productions based on innovations and the latest achievements of science.	BG 4 Vision - The transition to the circular economy will provide Bulgaria with economic growth, a clean environment, social well-being and a society with a high ecological awareness that thinks about future generations. The country's policy of transition to the circular economy will be established through a green and competitive economy, less waste and more resources and an economy in favor of consumers. As a result of the transformation of the economy, resource efficiency will be increased and the added value of industrial production. The consumption of some products will be replaced by services, and others will be accommodated for	BG 4.1 Objective(s) - 1. Green and competitive economy; 2. Less waste, more resources; 3. Economics in favor of consumers

1. Bioeconomic strategies for scientific research and innovation (bioeconomic strategies for scientific research and innovation) In Bulgaria, we have developed the Program "Scientific research, innovation and digitalization for intelligent transformation"		2. Social challenges, taking into account gender aspects and retraining needs (social challenges, taking into account gender aspects and retraining needs). Bulgaria has: A strategy for reducing poverty and especially for social inclusion with a horizon of 2030. NATIONAL STRATEGY FOR PROMOTING EQUALITY OF WOMEN AND MEN 2021-2030		3. finance and investment (finance and investment)		4. environment (environment. In Bulgaria imama: Strategy for transition to a circular economy, 2022 - 2027	
vision	objective(s)	vision	objective(s)	vision	objective(s)	vision	objective(s)
		factor for the realization of the rights of all citizens.	through the implementation of a unified, consistent and sustainable state policy			longer use. Individual productions will be linked so that they exist in symbiosis. The country will contribute to the provision of critical raw materials in the low union. The quantities that have been landfilled will be reduced to a minimum, and the rest will be put back into the production cycle or recycled. All this will be achieved by the society joining forces, and the state providing the necessary conditions and resources. The circular economy is a main and long-term priority of the country's development policy, for which the necessary human, material and financial resources will be mobilized.	

1. Bioeconomic strategies for scientific research and innovation (bioeconomic strategies for scientific research and innovation) In Bulgaria, we have developed the Program "Scientific research, innovation and digitalization for intelligent transformation"		2. Social challenges, taking into account gender aspects and retraining needs (social challenges, taking into account gender aspects and retraining needs). Bulgaria has: A strategy for reducing poverty and especially for social inclusion with a horizon of 2030. NATIONAL STRATEGY FOR PROMOTING EQUALITY OF WOMEN AND MEN 2021-2030		3. finance and investment (finance and investment)		4. environment (environment. In Bulgaria imama: Strategy for transition to a circular economy, 2022 - 2027	
vision	objective(s)	vision	objective(s)	vision	objective(s)	vision	objective(s)
	BG 1.2 Operational goals: Operational goal No. 1: Improving the research system and innovation performance of enterprises. With the realization of this goal, the ambition has reached levels of 70% compared to the EU, which will strengthen Bulgaria's position in the group of moderate innovators until 2027. Operational goal No. 2 Increasing the technological capacity of enterprises, increasing the environmental friendliness and internationalization of Bulgarian products and services. The emphasis in the implementation of goal No. 2 is on the thematic priority areas for intelligent specialization and technologies of Industry 4.0.8. Operational goal No. 3: Improving the capacity of human resources in the field of new technologies and innovations. Priority has been given to improving						

1. Bioeconomic strategies for scientific research and innovation (bioeconomic strategies for scientific research and innovation) In Bulgaria, we have developed the Program "Scientific research, innovation and digitalization for intelligent transformation"		2. Social challenges, taking into account gender aspects and retraining needs (social challenges, taking into account gender aspects and retraining needs). Bulgaria has: A strategy for reducing poverty and especially for social inclusion with a horizon of 2030. NATIONAL STRATEGY FOR PROMOTING EQUALITY OF WOMEN AND MEN 2021-2030		3. finance and investment (finance and investment)		4. environment (environment. In Bulgaria imama: Strategy for transition to a circular economy, 2022 - 2027	
vision	objective(s)	vision	objective(s)	vision	objective(s)	vision	objective(s)
	the overall environment for the development of skills and high-tech human resources in the thematic areas of smart specialization and Industry 4.0 and Industry 5.0 technologies.						

2. Croatia

1. bioeconomy strategies for research and innovation		2. social challenges, considering gender aspects and re-skilling needs		3. finance and investment		4. environment	
vision	objective(s)	vision	objective(s)	vision	objective(s)	vision	objective(s)
HR 1 Research and innovation is recognized and adequately supported in the National Bioeconomy Strategy as one of the keys areas that will contribute to developing bioeconomy in Croatia as dynamic and diversified activity, supporting the overall economic development.	HR 1.1 Encouraging scientific and research projects and their implementation in bioeconomy	HR 2 National Bioeconomy Strategies recognizes inclusivity and social equality, provides access to education and training, gives employment opportunities and ensures social well-being.	HR 2.1 Securing equal job opportunities for women	HR 3 National Bioeconomy Strategy secures sufficient funding from various sources to develop projects in bioeconomy and support additional investments by large companies and SMEs.	HR 3.1 Securing enough finances, by simplifying the procedures for national co-financing of the research and innovation projects in the area of bioeconomy	HR 4 Bioeconomy sector is in line with the sustainable goals and mitigates the impact of all climate changes as well as promotes principles of circularity.	HR 4.1 Supporting the projects in bioeconomy that lead to accomplishing climate neutrality and enhancing energy efficiency

1. bioeconomy strategies for research and innovation		2. social challenges, considering gender aspects and re-skilling needs		3. finance and investment		4. environment	
vision	objective(s)	vision	objective(s)	vision	objective(s)	vision	objective(s)
	HR 1.2 Improving and promoting cooperation between the scientific institutions/academia and private sector to develop joint projects		HR 2.2 Involving local communities into Strategy development to promote and understand the specific needs and challenges they face in respect to bioeconomy		HR 3.2 Defining priority projects for investments in the field of bioeconomy to foster further development		HR 4.2 Promoting bio-based and nature-based solutions in bioeconomy projects
	HR 1.3 Educating all stakeholders in bioeconomy value chain to gain better knowledge on research and innovation		HR 2.3 Securing work-life balance in new jobs created in bioeconomy		HR 3.3 Defining the objectives within the strategic national documents under which biodiversity projects can be funded		HR 4.3 Encouraging production of recyclable materials packaging and bio-based plastics
			HR 2.4 Developing education programs at the universities to foster the employment in bioeconomy		HR 3.4 Facilitating private investments into bioeconomy projects		HR 4.5 Implementing green procurement procedures in bioeconomy
			HR 2.5 Promoting bioeconomy jobs with the youth population				
			HR 2.6 Creating reskilling educational programs in bioeconomy as a measure to preventing the unemployment				

3. Czech Republic

1. bioeconomy strategies for research and innovation		2. social challenges, considering gender aspects and re-skilling needs		3. finance and investment		4. environment	
vision	objective(s)	vision	objective(s)	vision	objective(s)	vision	objective(s)
CZ 1 Creating an inclusive stable framework for cooperation and spin-off of research results enabling to include all possible stakeholders working in RDI. The basic conditions must be sustainability, cascade use of resources, increased use of bio-waste and available biomass, while remaining possibly economically profitable. With the help of innovation, digitalisation, new technology for traditional and new industry, RDI cooperation and education.	CZ 1.1 Mapping potential sources of available suitable biomass and identification of key stakeholders in the regions with respect to sustainable logistics and supply chains.	CZ 2 Balance between urban (as the main source of waste biomass) and rural space (as the main source of available biomass) with respect to land use and traditional activities, especially in rural areas. This situation could contribute to stopping rural depopulation by creating new jobs and business opportunities, especially with regard to gender and age equality.	CZ 2.1 Redefinition of certain wastes as usable biomass. Updating the waste sorting system already in place to take this into account	CZ 3 Opportunities for financing and investment in the circular bioeconomy at all levels and all types of key stakeholders with respect to sustainability, human resources, knowledge capital and technical infrastructure of the region. The source of financing should be state and EU funds as so as private investment.	CZ 3.1 Clear conditions for funding scientific research in circular bioeconomy (appropriate subsidy titles, long-term strategic documents guaranteeing continuity of the bioeconomy as a research topic).	CZ 4 Circular bioeconomy as a strengthening of sustainability between man and nature	CZ 4.1 Respect for the nature and integrity of the ecosystems to be used for biomass. Identification of (already protected) ecosystems that will not be used for commercial biomass extraction.
	CZ 1.2 Focusing research (if it is not already) on available sources of biomass and waste with a view to its future practical use in a circular bioeconomy.		CZ 2.2 Increasing public awareness of biomass cascading with respect to sustainability, human resources, knowledge capital and technical infrastructure of the region.				CZ 4.2 Newly established plants should (where possible) be exclusively oriented towards the circular bioeconomy.

1. bioeconomy strategies for research and innovation		2. social challenges, considering gender aspects and re-skilling needs		3. finance and investment		4. environment	
vision	objective(s)	vision	objective(s)	vision	objective(s)	vision	objective(s)
	CZ 1.3 Strengthen and/or create linkages between RDI and manufacturing with respect to technical infrastructure, human resources and knowledge capital in the different areas of manufacturing within the regions		CZ 2.3 Raise awareness of the mutual substitutability of current and circular bioeconomy products		CZ 3.3 Support of commodification of science and research results in the circular bioeconomy.		CZ 4.3 Implementation of the circular bioeconomy serves to meet most of the SDGs
	CZ 1.4 In the case of untapped biomass resources, proceed to its use strictly within the framework of the circular bioeconomy.		CZ 2.4 The education system will be able to respond to the demand for experts in the field of bioeconomy circularization in production, innovation and education		CZ 3.4 Financial support for formal and non-formal education in the field of circular bioeconomy		CZ 4.4 Consistent implementation of circular bioeconomy principles will contribute to the reduction of negative externalities
	CZ 1.5 Identification of legislative barriers and subsequent correction of the legal framework in order to use the available biomass in a desirable way		CZ 2.5 Food security should always take precedence over energy security in decision-making.		CZ 3.5 The creation of standards for bio-economic production (sustainability, circularity, etc.) that would make financial aid possible		
			CZ 2.6 New jobs in the circular bioeconomy should be supported through financial incentives and similar economic instruments from public budgets.		CZ 3.6 Investment incentives for investors in the circular bioeconomy		

1. bioeconomy strategies for research and innovation		2. social challenges, considering gender aspects and re-skilling needs		3. finance and investment		4. environment	
vision	objective(s)	vision	objective(s)	vision	objective(s)	vision	objective(s)
			CZ 2.7 The circular bioeconomy offers (at least partially) the possibility of maintaining living standards while reducing the burden on the environment		CZ. 3.7 The bioeconomy offers a comprehensive environmental solution. A possible source of funding for the circular bioeconomy could be the transfer of funds from subsidy titles that only address part of the issue.		

4. Greece

1. bioeconomy strategies for research and innovation		2. social challenges, considering gender aspects and re-skilling needs		3. finance and investment		4. environment	
vision	objective(s)	vision	objective(s)	vision	objective(s)	vision	objective(s)
GR 1 To establish Greece as a leader in the bioeconomy, fostering innovation, sustainable development, and environmental stewardship, while enhancing economic prosperity and improving the quality of life for its citizens.	GR 1.1 Invest in cutting-edge research to discover new bio-based technologies and processes.	GR 2 To build an inclusive and socially equitable bioeconomy in Greece, where gender diversity is embraced, and all individuals have access to education, training, and employment opportunities, fostering innovation and prosperity while ensuring social well-being.	GR 2.1 Implement policies and programs that address gender biases and stereotypes, fostering a supportive environment for women in bioeconomy-related fields.	GR 3 To establish Greece as a hub for sustainable finance and investment in the bioeconomy, fostering innovation, economic growth, and environmental conservation. Through strategic investments, Greece aims to drive the development of cutting-edge technologies, create jobs, and enhance the country's position as a leader in green and circular economy practices.	GR 3.1 Create a conducive regulatory environment that encourages domestic and foreign investment in bioeconomy sectors, offering incentives such as tax breaks, grants, and subsidies.	GR 4 Utilize natural resources efficiently, ensuring their preservation for future generations.	GR 4.1 Promote sustainable management of natural resources, including agricultural residues, forestry products, and marine resources.
	GR 1.2 Promote sustainable farming practices that optimize crop yields while preserving soil health and biodiversity.		GR 2.2 Encourage the participation of women in decision-making processes related to bioeconomy initiatives.		GR 3.2 Foster collaboration between the government, private sector, and research institutions through PPPs to leverage shared resources, knowledge, and funding for bioeconomy projects.		GR 4.2 Build a circular economy model, minimizing waste and maximizing resource efficiency.
	GR 1.3 Develop and promote circular economy models, encouraging the reuse, recycling, and upcycling of bio-based products and waste materials.		GR 2.3 Establish partnerships with educational institutions and industry players to ensure that curricula are aligned with the demands of the bioeconomy sector		GR 3.3 Support venture capital firms and angel investors specializing in green technologies and bioeconomy projects, ensuring a steady flow of funds into promising ventures.		GR 4.3 Enhance agricultural practices for higher yields and sustainability.

1. bioeconomy strategies for research and innovation		2. social challenges, considering gender aspects and re-skilling needs		3. finance and investment		4. environment	
vision	objective(s)	vision	objective(s)	vision	objective(s)	vision	objective(s)
	GR 1.4 Support startups and entrepreneurs working on bioeconomy-related projects through funding, mentorship, and access to resources.		GR 2.4 Engage with local communities to understand their specific needs and challenges related to the bioeconomy.		GR 3.4 Invest in infrastructure development, including research facilities, biorefineries, and waste management systems, to support the growth of bioeconomy sectors.		GR 4.4 Mitigate the impact of climate change through bio-based solutions.
	GR 1.5 Collaborate with the European Union and international organizations to align Greece's bioeconomy policies with global standards.		GR 2.5 Ensure that bioeconomy initiatives do not disproportionately affect vulnerable communities and work towards inclusive development.		GR 3.5 Collaborate with private investors to build and maintain a robust logistics and supply chain infrastructure for the seamless movement of bio-based products.		GR 4.5 Raise public awareness about the benefits and challenges of the bioeconomy.
	GR 1.6 Participate actively in international research initiatives and partnerships to stay at the forefront of bioeconomy innovation.		GR 2.6 Promote flexible work arrangements and family-friendly policies within the bioeconomy sector, ensuring a healthy work-life balance		GR 3.6 Collaborate with financial institutions and regulatory bodies to establish clear standards for green finance, ensuring transparency and accountability in bioeconomy-related investments.		GR 4.6 Develop a robust regulatory framework to ensure the sustainability and ethical practices within the bioeconomy.

5. Hungary

1. bioeconomy strategies for research and innovation		2. social challenges, considering gender aspects and re-skilling needs		3. finance and investment		4. environment	
vision	objective(s)	vision	objective(s)	vision	objective(s)	vision	objective(s)
HU 1 Creating a long-term bioeconomy strategy based on a multi-stakeholder and cross-sectoral approach, strategic cooperation, encouraging sustainable and circular management of bio-based resources, innovation, cascading use of biomass, and value-added extension of existing value chains.	HU 1.1 Mapping biomass potential, and relevant market players and conduct research on the interactions of main bioeconomy actors.	HU 2 Prioritization of inclusiveness in the economy in Hungary. Awareness of the public on the current challenges on natural resources and consumptions, cross-sectoral dialogue, promoting diversification of activities in rural areas, which can allow women, youth and marginalized groups to participate in training on bioeconomy, new value chains of biomass, and expanding employment opportunities.	HU 2.1 Strengthen the already existing knowledge, training programs and expertise on biomass use, such as biogas.	HU 3 Robust investment ecosystem supporting not just big but smaller scale ("small is beautiful") bioeconomy initiatives, startups and projects tailored to the specific potentials and needs of Hungary, for instance, the considerable biomass potential of Hungary and the enormous potential of use for biomethane and to remedy issues arising from local, national, regional and global issues such as the energy crisis.	HU 3.1 Attract private and public investment to support bio-based industries and research.	HU 4 Leveraging bioeconomic advancements to mitigate environmental impact and enhance ecosystem health, leading to a more resilient biodiversity in Hungary.	HU 4.1 To progressively shift towards renewable and sustainable resources (for instance, utilizing the significant biomethane potential of Hungary), minimizing reliance on non-renewable sources (gaining energy independence from other countries in the meantime) and mitigating environmental degradation while maintaining or ideally increasing land productivity (especially relevant for South-East Region of Hungary where most of our intensive agriculture takes place and where flooding and droughts are commonplace along with general soil degradation).
	HU 1.2 Bridge the gap between low (1-3) and high (7-9) Technology Readiness Levels (TRL) projects, allowing research opportunities to move up from low TRL (1-3) to medium (development TRL 4-6)		HU 2.2 Integrate in the curriculum in high education the topic of bioeconomy, linked to organic farming and better use of bio-based by-products		HU 3.2 Foster public-private partnerships to finance and scale bioeconomy ventures, particularly in promoting sectors like bioenergy, sustainable agriculture, and forestry.		HU 4.2 To safeguard and restore ecosystems (especially wetlands in the case of Hungary), fostering biodiversity and preserving natural habitats while taking advantage of nature-based solutions.

1. bioeconomy strategies for research and innovation		2. social challenges, considering gender aspects and re-skilling needs		3. finance and investment		4. environment	
vision	objective(s)	vision	objective(s)	vision	objective(s)	vision	objective(s)
	HU 1.3 The rational use of domestic biomass by cascading use (Bioeast concept paper for Hungary) HU 1.4 Paradigm shift and invigorate the policy -science dialogue through research and between experts, researchers, farmers and biomass producers.		HU 2.3 New forms of vocational training should consider the mobilization of experts, bringing opportunities to the rural areas should be prioritized when designing courses for farmers. HU 2.4 Raise awareness and target the youth, emphasizing awareness campaigns to increase the level of responsibility and commitment of the youth toward climate neutrality and circularity HU 2.5 As stated in the Concept Paper (BIOEAST) for Hungary, one of the priorities of the future bioeconomy strategy in Hungary will be "food first and healthy diet", which will support local organic farming while raising awareness on healthy diet in society. HU 2.6 Digitalization development in agriculture, including practice-oriented use of databases and, based on this the creation of programs for the development of digital skills		HU 3.3 Implement financial incentives and tax breaks to encourage investment in bioeconomy initiatives, ensuring a return on investment while contributing to societal and environmental well-being. HU 3.4 Create financial mechanisms and platforms that allow for easy access to funding, especially for startups and small to medium-sized enterprises venturing into bioeconomy innovations. HU 3.5 Increase transparency of green financing. Incentivize the banking sector about green financing, and diversity their investment to other areas besides energy efficiency HU 3.6 Support of small-scale, new generations of biorefineries, supporting local markets and finding a balance in the biomass energy mix.		HU 4.3 To contribute to mitigating climate change effects and adapting to the changing climate scenario. HU 4.4 To foster circular bioeconomy by minimizing waste and maximizing resource use efficiency.

6. Poland

1. bioeconomy strategies for research and innovation		2. social challenges, considering gender aspects and re-skilling needs		3. finance and investment		4. environment	
vision	objective(s)	vision	objective(s)	vision	objective(s)	vision	objective(s)
PL 1 Poland leads Central and Eastern Europe by harnessing the potential of science and innovation in developing a sustainable and circular bioeconomy, based on: (1) increased productivity and cascaded use of bio-waste from bioeconomy sectors (2) development of novel and advanced technologies (3) enhanced business-policy makers-science cooperation and development of process innovations	PL 1.1 Increasing the agricultural productivity through sustainable intensification	PL 2 A modern and responsible society, enabling the development of all segments of society and tackling demographic challenges, climate change and the use of its renewable resource potential	PL 2.1 Significantly strengthen the relationship between business, science and educational activities in the field of sustainability and climate change, as well as in the search for novel technological, organizational and social solutions	PL 3 Self-financing circular and sustainable bioeconomy at regional and national level with effective use of available opportunities and possibilities	PL 3.1 Increasing access to finance and advancing the level of public-private endeavor in pooling resources for R&I	PL 4 A low-carbon economy based on sustainable and circular value chains, with biodiversity and natural resources preserved and not degraded for future generations	PL 4.1 Transforming the national economy into a resource-efficient, green and competitive low-carbon economy
	PL 1.2 Cascading the use of agricultural and forest residues potential (increasing circularity) and the added value of biomass through innovative bio-products and technologies		PL 2.2 Carrying out systematic and meaningful education activities on sustainability and climate change, waste management, responsible consumption and healthy diet				
	PL 1.3 Increasing novel and advanced bio-refinery technologies and products		PL 2.3 Increasing social awareness of the production and use of biogas and biomethane		PL 3.2 Accelerating sector's technological modernization - developing integrated resource-based strategies		PL 4.2 Developing fossil fuel substitution pathways
					PL 3.3 Increasing the investment in R&D in bioeconomy sectors		PL 4.3 Accelerating the energy transition while taking care of biodiversity, natural resources (e.g. water resources)

1. bioeconomy strategies for research and innovation		2. social challenges, considering gender aspects and re-skilling needs		3. finance and investment		4. environment	
vision	objective(s)	vision	objective(s)	vision	objective(s)	vision	objective(s)
	PL 1.4 Strengthening significantly the relationship between business and science and educational activities in the field of sustainability and climate change		PL 2.4 Transformation and re-skilling of employees in mining (e.g. coal) and related sectors		PL 3.4 Increasing the application for funding under the European Research Area by universities, institutes and companies.		4.4 Development of the 4th generation biogas and biomethane sector, taking into account emission reduction and climate change potentials
	PL 1.5 Achieving coherent and planned research funding, based on long-term strategic planning.		PL 2.5 Ensuring women have equal opportunities to build a shared, sustainable future (Women must have equal access to both work on the factory floor and in high-tech laboratories)		PL 3.5 Strengthening infrastructure for production plants and industrial processes conversion and pilot test areas to increase the level of technological readiness.		PL 4.5 Increasing the share of renewable energy at national level
	PL 1.6 Increasing understanding of the bioeconomy at business, scientific, governmental and consumer levels, better use of the education system at all levels.		PL 2.6 Tackling demographic and social changes related to an ageing population, internal and external migration of young people - affecting various aspects of economic and social development in the countryside and slowdown in generational change in sectors of bioeconomy e.g. agriculture or fisheries (Need to Adopt National/Regional Bioeconomy Strategies - Negative Effects of Ageing and Demographics)		PL 3.6 Building and leveraging the first of its kind bioeconomy entrepreneurship ecosystem, supported by public and private funding institutions		PL 4.6 Sustainable water management, including ensuring access to clean water for the public and the economy and achieving good water status

1. bioeconomy strategies for research and innovation		2. social challenges, considering gender aspects and re-skilling needs		3. finance and investment		4. environment	
vision	objective(s)	vision	objective(s)	vision	objective(s)	vision	objective(s)
	PL 1.7 Developing programs to raise awareness and educate primary sectors producers on the principles and elements of a circular economy.		PL 2.7 Introducing changes in the socio-economic model to enable the transition to a sustainable circular bioeconomy				
	PL 1.8 Increase cooperation and development of evidence-based policies including R&I aspects		PL 2.8 Increasing the public awareness of the results of works and projects financed with payments for the ecosystem services				

7. Romania

1. bioeconomy strategies for research and innovation		2. social challenges, considering gender aspects and re-skilling needs		3. finance and investment		4. environment	
vision	objective(s)	vision	objective(s)	vision	objective(s)	vision	objective(s)
RO 1 Create a more dynamic innovation ecosystem for supporting bioeconomy development and innovation. Strengthen the RDI capacity, considering a holistic and interdisciplinary approach.	RO 1.1 Raising the level of knowledge and understanding of the principles of bio-economy, circular economy and sustainable development.	RO 2 Increase the social acceptance of bioeconomy transition and jointly develop/shape a sustainable, inclusive and resilient bioeconomy	RO 2.1 education and training, capacity building to ensure that workers and entrepreneurs adapt to new bio-markets and new bio-related technologies	RO 3 Increase funding and finance for the bioeconomy by a coherent approach, develop understandable and accessible bioeconomy-targeted funding and financial mechanisms	RO 3.1 Awareness raising about financing mechanisms. Support and comprehensive technical assistance related to financial instruments.	RO 4 Sustainable use of bioresources, nature conservation and limiting negative impacts on the environment make the economy more sustainable and in harmony with nature conservation and environmental protection	RO 4.1 efficient management of natural resources
	RO 1.2 Enhance the cooperation between science and business Improving cooperation between research and Industry		RO 2.2 capacity building along the entire value chain		RO 3.2 create synergies and complementarities between funds		RO 4.2 focus on the cascading use of biomass to provide products with high added value
	RO 1.3 Facilitate awareness raising, best practices and capacity building		RO 2.3 ensure life-long learning		RO 3.3 Integrated projects: private and public stakeholders (e.g. developing integrated plans in rural communities taking into consideration social-economic-environment dimensions)		RO 4.3 circular and value-added use of biomass
	RO 1.4 integrating actors across value chains to experiment and work together on new solutions that will provide enhanced sustainability and circularity, and which are adapted to local condition		RO 2.4 create green jobs, special attention to rural communities		RO 3.4 Medium and large-scale business cases supporting private BE infrastructure;		

1. bioeconomy strategies for research and innovation		2. social challenges, considering gender aspects and re-skilling needs		3. finance and investment		4. environment	
vision	objective(s)	vision	objective(s)	vision	objective(s)	vision	objective(s)
	RO 1.5 strengthen cross-sectorial and holistic perspectives		RO 2.5 youth education, stronger involvement		RO 3.5 Small scale business cases individual projects;		
	RO 1.6 strengthen the competitiveness of Bioeconomy Clusters and Innovation Ecosystems				RO 3.6 encourage and initiate new ways of financing, launch innovative funding schemes		

8. Serbia

Summary of sent visions and objectives

1. bioeconomy strategies for research and innovation		2. social challenges, considering gender aspects and re-skilling needs		3. finance and investment		4. environment	
vision	objective(s)	vision	objective(s)	vision	objective(s)	vision	objective(s)
RS 1 The vision is primarily concerned with agriculture as an important sector of the Serbian economy. Bio-economic innovations and new technologies should enable better use of resources and in general the application of the sustainable model of complete cycle circularity) in agriculture. Another focus is on education- they want to create Study programs new modules on bioeconomy within the study programs of the Faculty of Agriculture in the Republic of Serbia.	RS 1.1 Including bioeconomy into Strategy of Scientific and Technological Development of the Republic of Serbia	RS 2 Make bioeconomy strategies in forestry, aquaculture and fisheries environment where women and men are equal participants in the bioeconomy sectors. This includes equal access to resources, education and business opportunities. Education and retraining are key elements of the strategy. Job creation.	RS 2.1 a) to ensure that women and men have equal opportunities in the forestry sector	RS 3 The bioeconomy strategy (from the aspect of forestry sector) relies on different sources of financing. Establish system of sustainable financing, Ensuring Investment system through EU funds and national funds	RS 3.1 ad a) Allocate funds from the state, provincial and local self-government budgets to support the implementation of the Bioeconomy Strategy Use of international funds (e.g. IPA, Interreg) and EU funds (e.g. Horizon Europe) The private sector, including forestry companies and the wood industry, can participate in the financing	RS 4 General objective - Conserved and improved state of the environment and circular economy What to support - bioeconomy practices and sustainable management in forestry and agriculture. With the help of defined of special fees and charges for activities aimed at preserving the environment would contribute to the creation of conditions for achieving positive effects of the agriculture-environment feedback loop.	RS 4.1 Development of sustainable products and innovations
	RS 1.2 Wide application of digitization		RS 2.2 b) Given the changes in technology and market demands, continuous training and (eventual) retraining of the workforce in the forestry sector is required		RS 3.2 Increasing the export and trade of forest products can provide additional revenues that can be used to finance the Bioeconomy Strategy. This includes the export of wood products, biomass, non-wood forest products, etc.		RS 4.2 Development of rural areas

1. bioeconomy strategies for research and innovation		2. social challenges, considering gender aspects and re-skilling needs		3. finance and investment		4. environment	
vision	objective(s)	vision	objective(s)	vision	objective(s)	vision	objective(s)
	RS 1.3 Education and training of people in the forestry sector		RS 2.3 c) Encouraging education and promotion of forestry among women and young people		RS 3.3 Ad b) Where finance should go- projects on bioeconomy and environment- a clear correlation between the existing green credits and incentives and projects that have a bioeconomic character and make the necessary improvements to the environment for investments and financing.		RS 4.3 Circular use of wood products
	RS 1.4 cooperation between the government, the private sector, educational and research organizations and local communities		RS 2.4 Previously prepared bioeconomy strategy in agriculture in the Republic of Serbia				RS 4.4 Forest biomass as a substitute for fossil fuels
	RS 1.5 Identification of potential for bioeconomy		RS 2.5 Employed professional and qualified workforce.				RS 4.5 Climate change management (e.g. forests play a key role in absorbing carbon from the atmosphere, reducing the greenhouse effect)
	RS 1.6 Research related to increasing the efficiency of recycling		RS2.6 Identification of the potential of bioeconomy actors and the possibility of implementing the				RS 4.6 Pollution reduction, waste management, nature protection
	RS 1.7 Greater funds and resources allocated to research and innovation		RS 2.7 Raising awareness of the importance of bioeconomy and circular economy				RS 4.7 Revitalization and preservation of natural resources - soil, water, etc., restoration of ecosystems, sustainability of biodiversity ...

1. bioeconomy strategies for research and innovation		2. social challenges, considering gender aspects and re-skilling needs		3. finance and investment		4. environment	
vision	objective(s)	vision	objective(s)	vision	objective(s)	vision	objective(s)
	RS 1.8 Improvement of legal frameworks relied on above.						RS 4.8 Raising awareness of the importance of bioeconomy and circular economy
							RS 4.9 Reducing pressure on protected natural areas

9. Slovakia

1. bioeconomy strategies for research and innovation		2. social challenges, considering gender aspects and re-skilling needs		3. finance and investment		4. environment	
vision	objective(s)	vision	objective(s)	vision	objective(s)	vision	objective(s)
SK 1 To achieve sustainable and resilient bioeconomy through cutting-edge research and innovation	SK 1.1 Increase research and innovation capacity and excellence in bioeconomy-related areas	SK 2 To achieve a dynamic and socially inclusive bioeconomy that brings benefits to all segments of society and ensures a sustainable legacy for generations to come	SK 2.1 Change attitudes of society towards sustainable use of resources	SK 3 To catalyze strategic financing of bioeconomy by attracting public and private investments that support research, innovation and implementation of bio-based solutions	SK 3.1 Adopt financial strategies and tools that prioritize impact, inclusivity and long-term value creation	SK 4 To position bioeconomy as a catalyst of positive climate action and key driver of achieving sustainability	SK 4.1 Foster practices that contribute to implementing the principles of circular bioeconomy with the aim of minimizing waste, reducing greenhouse gas emissions and designing added-value products with a lifecycle approach
	SK 1.2 Create supportive policy framework providing incentives for R&D, incl. funding for infrastructure		SK 2.2 Develop educational programs and training opportunities in bioeconomy for all generations				
	SK 1.3 Establish funding mechanisms, grants and incentives to support research projects that contribute to the growth and competitiveness of bioeconomy		SK 2.3 Prioritize education and capacity building to develop skilled workforce capable of driving innovation in bioeconomy				
					SK 3.2 Foster collaboration between public and private sectors through partnerships that leverage both financial and non-financial resources for the advancement of bioeconomy		SK 4.2 Mitigate environmental challenges through innovative technologies and responsible resource management, while harmonizing environmental, economic and social balance
					SK 3.3 Increase the income of farmers and foresters by increasing the added value of bio-based products, services and technologies, creating new sources of income not only for farmers but for the entire value chains and for the country		
					SK 3.4 Provide financial support and incentives for start-ups and SMEs in bioeconomy		

10. Slovenia

1. bioeconomy strategies for research and innovation		2. social challenges, considering gender aspects and re-skilling needs		3. finance and investment		4. environment	
vision	objective(s)	vision	objective(s)	vision	objective(s)	vision	objective(s)
SI 1 Future RDI in the field of bioeconomy to be oriented at: Strategic Collaboration and Networking Vision. Creating a Connected and Collaborative Bioeconomy Ecosystem (3-5 years). Research and Education Vision Fostering a Knowledge-Driven Bioeconomy (10 years): Transforming Traditional Sectors through Biorefining and Technology Integration Vision (10-20 years) Sustainability and Circular Economy Vision (20 years)	SI 1.1 Promote Small-Scale Biorefineries Innovation in SMEs: agri-food companies both economic growth and environmental sustainability.	SI 2 Concentrated on education and Cooperation: Youth Engagement and Education Vision (5-8 years): Cultivating the Next Generation of Bioeconomy Leaders. Interdisciplinary Collaboration Vision (5-7 years): Fostering Synergy Across Disciplines and Ministries for a Unified Bioeconomy. Professional Development and Knowledge Expansion Vision (5-7 years): Building a Knowledge-Driven Workforce for a Sustainable Bioeconomy	SI 2.1 Youth Engagement: Ensure the inclusion of young minds in the bioeconomy transformation. New Opportunities for the Youth: <i>Launch internships, scholarships, and innovation competitions.</i> Collaborate with universities to provide hands-on experiences in the sector.	SI 3 Creation of financial Framework for: Actionable Investment Strategies and Case Studies Vision Setting Precedents for Successful Bioeconomy Investments (1-3 years), Funding Infrastructure and Collaborative Investments Vision Cultivating Collaborative and streamlined Bioeconomy Investments (3-5 years), Investment Attractiveness and Communication Vision Propelling Bioeconomy with Profitable Investment Opportunities (5-7 years)	SI 3.1 Highlight the profitability and ROI of new technologies in the bioeconomy sector.	SI 4 With help of new research approaches and innovations to reduce the negative environmental impacts	SI 4.1 Reduction of Carbon Footprint: Achieve a significant reduction in carbon emissions through the adoption of bioeconomy principles.

1. bioeconomy strategies for research and innovation		2. social challenges, considering gender aspects and re-skilling needs		3. finance and investment		4. environment	
vision	objective(s)	vision	objective(s)	vision	objective(s)	vision	objective(s)
	SI 1.2 Establish Strong Core Companies in conventional bioeconomy sectors like sawmills, sugar factories, or mills to facilitate large-scale industrial biomass collection. Encouraging companies to innovating and establishing strong vertical systems. Engage Large Industry Players: Recognize and utilize their potential to influence government decisions for a smoother transition to the circular bioeconomy.		SI 2.2 Knowledge Centers: An establishment of "Knowledge Centers" linked to companies with the outcome of trained future employees and the initiation of start-ups.		SI 3.2 Create Open-Access Pilot Infrastructure: Establish a conducive environment for pilot infrastructures, ensuring broader access beyond private companies and addressing commercial limitations. Launch pilot biorefineries, especially in sectors like bakery, bread-making, and alternative proteins.		SI 4.2 Waste Minimization: Achieve near-zero waste in agricultural and industrial processes through biorefineries and circular value chains.
	SI 1.3 Expand Regional and International Collaboration Digitalization: Utilize e-solutions to streamline collaboration and knowledge sharing. Facilitate Knowledge Transfer from Research Institutions: to companies to collaborate with research organizations,		SI 2.3 Broaden Educational Perspectives: Re-evaluate and redesign educational curricula to provide a more holistic view of interconnected disciplines, ensuring students aren't overly specialized and isolated from broader subjects, especially pivotal areas like the bioeconomy.		SI 3.3 Private Funding: <i>Invite Investors to a discussion on how to promote</i> a pipeline of investable projects in collaboration: ALFI, Pečečnik, Pipistrel, SPS.		SI 4.3 Energy Self-Sufficiency: Attain a higher degree of energy self-sufficiency using renewable sources such as biomass.

1. bioeconomy strategies for research and innovation		2. social challenges, considering gender aspects and re-skilling needs		3. finance and investment		4. environment	
vision	objective(s)	vision	objective(s)	vision	objective(s)	vision	objective(s)
	SI 1.4 Establish a Bioeconomy Committee: of experts		SI 2.4 Foster Inter-Ministerial Collaboration: Encourage closer collaboration between ministries to eliminate the uncoordinated approach and defensive advancement strategies. Empower Civil Servants with Bioeconomy Knowledge:		SI 3.4 Involvement of Banks: Invite banks with green bonds on a discussion on how to promote a pipeline of bankable projects in collaboration: SID, NLB, Intesa Sanpaolo Bank.		SI 4.4 Climate Resilience: Develop climate-resilient agricultural practices, safeguarding food security and rural livelihoods
	SI 1.5 National Bioeconomy Hub's primary role is to facilitate connections and networking across different sectors and stakeholders, rather than primarily focusing on technical knowledge. Ensure active participation from industry players in the hub's initiatives.		SI 2.5 Simplify Legal and Permit Procedures: Expedite the process for companies to obtain essential environmental and spatial permits, reducing bureaucratic delays that hinder the bioeconomy projects.		SI 3.5 Enhance Investment Knowledge: Promote awareness and training on modern investment collaboration models like Joint Ventures. Reevaluate and clarify existing partnership stipulations to ensure transparency, scalability, and increased motivation from the private sector.		
	SI 1.6 Establish a Bioeconomy Strategy: Develop and implement a comprehensive national Bioeconomy strategy for Slovenia, aligning with best practices from leading EU nations		SI 2.6 Address Political Bottlenecks: Identify and rectify the lack of political support that maintains the agricultural status quo, hindering progress in sustainable practices.		SI 3.6 Partnerships: Facilitate connections between businesses looking for partnerships.		

1. bioeconomy strategies for research and innovation		2. social challenges, considering gender aspects and re-skilling needs		3. finance and investment		4. environment	
vision	objective(s)	vision	objective(s)	vision	objective(s)	vision	objective(s)
	SI 1.7 Recognise and adapt best practices from the automotive industry in Slovenia, as they exhibit successful vertically connected systems and a rapid move towards bio-innovation.		SI 2.7 Boost Bioeconomy Knowledge and Training: While bioeconomy is still a new field in Slovenia, promote its understanding through round tables, events, and training sessions, aiming for a carbon-neutral footprint by 2050.		SI 3.7 Revamp Funding Schemes: Overhaul the national public funding structure to be more responsive to current challenges. Streamline the bureaucratic processes, foster inter-ministerial collaboration, and increase project greenlighting to motivate and involve the private sector more effectively.		
	SI 1.8 Engage Traditional Sectors: Rebrand and rejuvenate traditional sectors like forestry and agriculture through innovation.		SI 2.8 Promote Interdisciplinary Awareness: Cultivate an environment where professionals, even if specialized, are made aware of adjacent fields, fostering a deeper understanding and appreciation of interconnected topics.		SI 3.8 Promote Decisive Action: Redirect focus from trivial debates to initiate bold and impactful steps that will elevate the circular bioeconomy in Slovenia, prioritizing substantial investments in the sector.		

1. bioeconomy strategies for research and innovation		2. social challenges, considering gender aspects and re-skilling needs		3. finance and investment		4. environment	
vision	objective(s)	vision	objective(s)	vision	objective(s)	vision	objective(s)
	SI 1.9 Organize Niche Value Chain Workshops: Focus on specific value chains, for successful circular solution implementation, leading to either research project applications or investment plans.		SI 2.9 Prioritize Personnel Development: Identify and bridge knowledge gaps among employees based on anticipated industry trends over the next 5 to 10 years. Design and implement training programs accordingly. Provide Specialized Training: Instead of generic CSR trainings, offer specific training in areas like carbon footprint relevant to the company's industry. Ensure that such trainings are solution-oriented and address real challenges faced by the company.		SI 3.9 Real Life Cases: Provide Case studies on investments for deploying circular bioeconomy value chains with business cases.		
	SI 1.10 Biomass Flow Optimisation: Identify and valorise optimal biomass flows and hopots.						

Annex 2: List of detail objectives

CEE2ACT WP 6. 1. OBJECTIVES

1. Bioeconomy strategies for research and innovation

1. BETTER USE OF NATURAL RESOURCES CZ 1.4 In the case of untapped biomass resources, proceed to its use strictly within the framework of the circular bioeconomy.
2. BETTER USE OF NATURAL RESOURCES HU 1.3 The rational use of domestic biomass by cascading use (Bioeast concept paper for Hungary)
3. BETTER USE OF NATURAL RESOURCES PL 1.2 Cascading the use of agricultural and forest residues potential (increasing circularity) and the added value of biomass through innovative bio-products and technologies
4. BETTER USE OF NATURAL RESOURCES RS 1.2 Wide application of digitization
5. BETTER USE OF NATURAL RESOURCES SI 1.10 Biomass Flow Optimization: Identify and valorise optimal biomass flows and hotspots.
6. COOPERATION CZ 1.3 Strengthen and/or create linkages between RDI and manufacturing with respect to technical infrastructure, human resources and knowledge capital in the different areas of manufacturing within the regions
7. COOPERATION RS 1.4 cooperation between the government, the private sector, educational and research organizations and local communities
8. COOPERATION GR 1.5 Collaborate with the European Union and international organizations to align Greece's bioeconomy policies with global standards.
9. COOPERATION GR 1.6 Participate actively in international research initiatives and partnerships to stay at the forefront of bioeconomy innovation.
10. COOPERATION POLICY PL 1.8 Increase cooperation and development of evidence-based policies including R&I aspects
11. COOPERATION RO 1.2 Enhance the cooperation between science and business Improving cooperation between research and Industry
12. COOPERATION RO 1.4 integrating actors across value chains to experiment and work together on new solutions that will provide enhanced sustainability and circularity, and which are adapted to local condition
13. COOPERATION RO 1.5 strengthen cross-sectorial and holistic perspectives
14. COOPERATION SI 1.2 Establish Strong Core Companies in conventional bioeconomy sectors like sawmills, sugar factories, or mills to facilitate large-scale industrial biomass collection. Encouraging companies to innovating and establishing strong vertical systems. Engage Large Industry Players: Recognize and utilize their potential to influence government decisions for a smoother transition to the circular bioeconomy.
15. COOPERATION SI 1.3 Expand Regional and International Collaboration Digitalization: Utilize e-solutions to streamline collaboration and knowledge sharing.. Facilitate Knowledge Transfer from Research Institutions: to companies to collaborate with research organizations,
16. COOPERATION SI 1.4 Establish a Bioeconomy Committee: of experts
17. COOPERATION SI 1.5 National Bioeconomy Hub's primary role is to facilitate connections and networking across different sectors and stakeholders, rather than primarily focusing on technical knowledge. Ensure active participation from industry players in the hub's initiatives.
18. COOPERATION SI 1.8 Engage Traditional Sectors: Rebrand and rejuvenate traditional sectors like forestry and agriculture through innovation.

19. COOPERATION EDUCATION SI 1.9 Organize Niche Value Chain Workshops: Focus on specific value chains, for successful circular solution implementation, leading to either research project applications or investment plans.
20. COOPERATION EDUCATION HR 1.2 Improving and promoting cooperation between the scientific institutions/academia and private sector to develop joint projects
21. COOPERATION EDUCATION PL 1.4 Strengthening significantly the relationship between business and science and educational activities in the field of sustainability and climate change
22. COOPERATION POLICY PL 1.8 Increase cooperation and development of evidence-based policies including R&I aspects
23. ECONOMIC RO 1.6 strengthen the competitiveness of Bioeconomy Clusters and Innovation Ecosystems
24. ECONOMIC SI 1.1 Promote Small-Scale Biorefineries Innovation in SMEs: agri-food companies both economic growth and environmental sustainability.
25. ECONOMIC PL 1.1 Increasing the agricultural productivity through sustainable intensification
26. ECONOMIC COOPERATION BG 1.4 Operational goal No. 2 Increasing the technological capacity of enterprises, increasing the environmental friendliness and internationalization of Bulgarian products and services. The emphasis in the implementation of goal No. 2 is on the thematic priority areas for intelligent specialization and technologies of Industry 4.0.8
27. ECONOMIC EDUCATION BG 1.5 Operational goal No. 3: Improving the capacity of human resources in the field of new technologies and innovations. Priority has been given to improving the overall environment for the development of skills and high-tech human resources in the thematic areas of smart specialization and Industry 4.0 and Industry 5.0 technologies.
28. ECONOMIC RESEARCH BG 1.1 Goal/s Strategic goals:1. To develop and position Bulgaria as a center of medium- and high-tech innovations in the strategic areas in which the country has established capacity and market positions, as well as to recognize the competence to compete on the world market, increasing the country's national and regional innovation performance;
29. EDUCATION PL 1.6 Increasing understanding of the bioeconomy at business, scientific, governmental and consumer levels, better use of the education system at all levels.
30. EDUCATION PL 1.7 Developing programmes to raise awareness and educate primary sectors producers on the principles and elements of a circular economy.
31. EDUCATION RO 1.1 Raising the level of knowledge and understanding of the principles of bioeconomy, circular economy and sustainable development.
32. EDUCATION RO 1.3 Facilitate awareness raising, best practices and capacity building
33. EDUCATION RS 1.3 Education and training of people in the forestry sector
34. EDUCATION RESEARCH HR 1.3 Educating all stakeholders in bioeconomy value chain to gain better knowledge on research and innovation
35. EDUCATION BETTER USE OF NATURAL RESOURCES GR 1.2 Promote sustainable farming practices that optimize crop yields while preserving soil health and biodiversity.
36. FUNDING GR 1.4 Support startups and entrepreneurs working on bioeconomy-related projects through funding, mentorship, and access to resources.
37. FUNDING POLICY PL 1.5 Achieving coherent and planned research funding, based on long-term strategic planning.
38. FUNDING POLICY SK 1.2 Create supportive policy framework providing incentives for R&D, incl. funding for infrastructure

39. FUNDING POLICY SK 1.3 Establish funding mechanisms, grants and incentives to support research projects that contribute to the growth and competitiveness of bioeconomy
40. POLICY CZ 1.5 Identification of legislative barriers and subsequent correction of the legal framework in order to use the available biomass in a desirable way
41. POLICY RS 1.1 Including bioeconomy into Strategy of Scientific and Technological Development of the Republic of Serbia
42. POLICY RS 1.8 Improvement of legal frameworks relied on above
43. POLICY SI 1.6 Establish a Bioeconomy Strategy: Develop and implement a comprehensive national Bioeconomy strategy for Slovenia, aligning with best practices from leading EU nations
44. POLICY COOPERATION HU 1.4 Paradigm shift and invigorate the policy -science dialogue through research and between experts, researchers, farmers and biomass producers.
45. RESEARCH GR 1.1 Invest in cutting-edge research to discover new bio-based technologies and processes.
46. RESEARCH HU 1.2 Bridge the gap between low (1-3) and high (7-9) Technology Readiness Levels (TRL) projects, allowing research opportunities to move up from low TRL (1-3) to medium (development TRL 4-6)
47. RESEARCH RS 1.5 Identification of potential for bioeconomy
48. RESEARCH RS 1.6 Research related to increasing the efficiency of recycling
FUNDING RS 1.7 Greater funds and resources allocated to research and innovation
RESEARCH CZ 1.1 Mapping potential sources of available suitable biomass and identification of key stakeholders in the regions with respect to sustainable logistics and supply chains.
49. RESEARCH CZ 1.2 Focusing research (if it is not already) on available sources of biomass and waste with a view to its future practical use in a circular bioeconomy.
50. RESEARCH BG 1.2. To support the deployment and establishment of a sustainable, modern, dynamic, inclusive, data-driven and globally connected research, innovation and entrepreneurship ecosystem in Bulgaria.
51. RESEARCH SK 1.1 Increase research and innovation capacity and excellence in bioeconomy-related areas
52. RESEARCH COOPERATION HR 1.1 Encouraging scientific and research projects and their implementation in bioeconomy
53. RESEARCH COOPERATION HU 1.1 Mapping biomass potential, and relevant market players and conduct research on the interactions of main bioeconomy actors.
54. RESEARCH BETTER USE OF NATURAL RESOURCES GR 1.3 Develop and promote circular economy models, encouraging the reuse, recycling, and upcycling of bio-based products and waste materials.
55. RESEARCH ECONOMIC BG 1.3 Operational goals: Operational goal No. 1: Improving the research system and innovation performance of enterprises. With the realization of this goal, the ambition has reached levels of 70% compared to the EU, which will strengthen Bulgaria's position in the group of moderate innovators until 2027.
56. RESEARCH ECONOMIC PL 1.3 Increasing novel and advanced bio-refinery technologies and products
57. RESEARCH EDUCATION SI 1.7 Recognize and adapt best practices from the automotive industry in Slovenia, as they exhibit successful vertically connected systems and a rapid move towards bio-innovation.

2. social challenges, considering gender aspects and re-skilling needs

1. COOPERATION GR 2.4 Engage with local communities to understand their specific needs and challenges related to the bioeconomy.
2. COOPERATION GR 2.3 Establish partnerships with educational institutions and industry players to ensure that curricula are aligned with the demands of the bioeconomy sector
3. COOPERATION PL 2.1 Significantly strengthen the relationship between business, science and educational activities in the field of sustainability and climate change, as well as in the search for novel technological, organizational and social solutions
4. COOPERATION RS 2.4 Previously prepared bioeconomy strategy in agriculture in the Republic of Serbia
5. COOPERATION RS2.6 Identification of the potential of bioeconomy actors and the possibility of implementing the
6. COOPERATION SI 2.4 Foster Inter-Ministerial Collaboration: Encourage closer collaboration between ministries to eliminate the uncoordinated approach and defensive advancement strategies. Empower Civil Servants with Bioeconomy Knowledge:
7. COOPERATION ECONOMIC POLICY EQUITY BG 2.1 - The limited and inefficient cooperation with research organizations, higher education institutions and business, leading to low levels of technology and knowledge transfer and for the commercialization of the results of R&D and innovation in industry and – accordingly – in a slow pace between the production of products with high added value and implementing solutions to address social and economic challenges. The strategic goal of the National Strategy for Promoting the Equality of Women and Men for 2021-2030 is to contribute to the achievement of de facto equality of women and men in Bulgaria, through the implementation of a unified, consistent and sustainable state policy
8. COOPERATION EDUCATION HR 2.2 Involving local communities into Strategy development to promote and understand the specific needs and challenges they face in respect to bioeconomy
9. ECONOMIC CZ 2.7 The circular bioeconomy offers (at least partially) the possibility of maintaining living standards while reducing the burden on the environment
10. ECONOMIC RO 2.2 capacity building along the entire value chain
11. EDUCATION HR 2.4 Developing education programs at the universities to foster the employment in bioeconomy
12. EDUCATION HU 2.1 Strengthen the already existing knowledge, training programs and expertise on biomass use, such as biogas.
13. EDUCATION HU 2.2 Integrate in the curriculum in high education the topic of bioeconomy, linked to organic farming and better use of bio-based by-products
14. EDUCATION HU 2.4 Raise awareness and target the youth, emphasizing awareness campaigns to increase the level of responsibility and commitment of the youth toward climate neutrality and circularity
15. EDUCATION HU 2.6 Digitalization development in agriculture, including practice-oriented use of databases and, based on this the creation of programs for the development of digital skills
16. EDUCATION PL 2.2 Carrying out systematic and meaningful education activities on sustainability and climate change, waste management, responsible consumption and healthy diet
17. EDUCATION PL 2.3 Increasing social awareness of the production and use of biogas and biomethane
18. EDUCATION PL 2.7 Introducing changes in the socio-economic model to enable the transition to a sustainable circular bioeconomy

19. EDUCATION PL 2.8 Increasing the public awareness of the results of works and projects financed with payments for the ecosystem services
20. EDUCATION SI 2.2 Knowledge Centers: An establishment of "Knowledge Centers" linked to companies with the outcome of trained future employees and the initiation of start-ups.
21. EDUCATION SI 2.3 Broaden Educational Perspectives: Re-evaluate and redesign educational curricula to provide a more holistic view of interconnected disciplines, ensuring students aren't overly specialized and isolated from broader subjects, especially pivotal areas like the bioeconomy.
22. EDUCATION RO 2.3 ensure life-long learning
23. EDUCATION RO 2.5 youth education, stronger involvement
24. EDUCATION RS 2.3 c) Encouraging education and promotion of forestry among women and young people
25. EDUCATION RS 2.7 Raising awareness of the importance of bioeconomy and circular economy
26. EDUCATION SI 2.7 Boost Bioeconomy Knowledge and Training: While bioeconomy is still a new field in Slovenia, promote its understanding through round tables, events, and training sessions, aiming for a carbon-neutral footprint by 2050.
27. EDUCATION SK 2.1 Change attitudes of society towards sustainable use of resources
28. EDUCATION SK 2.2 Develop educational programs and training opportunities in bioeconomy for all generations
29. EDUCATION COOPERATION SI 2.8 Promote Interdisciplinary Awareness: Cultivate an environment where professionals, even if specialized, are made aware of adjacent fields, fostering a deeper understanding and appreciation of interconnected topics.
30. EDUCATION CZ 2.1 Redefinition of certain wastes as usable biomass. Updating the waste sorting system already in place to take this into account
31. EDUCATION CZ 2.2 Increasing public awareness of biomass cascading with respect to sustainability, human resources, knowledge capital and technical infrastructure of the region.
32. EDUCATION CZ 2.3 Raise awareness of the mutual substitutability of current and circular bioeconomy products
33. EDUCATION CZ 2.4 The education system will be able to respond to the demand for experts in the field of bioeconomy circularization in production, innovation and education
34. EDUCATION COOPERATION HU 2.3 New forms of vocational training should consider the mobilization of experts, bringing opportunities to the rural areas should be prioritized when designing courses for farmers.
35. EDUCATION EMPLOYMENT RO 2.1 education and training, capacity building to ensure that workers and entrepreneurs adapt to new bio-markets and new bio-related technologies
36. EDUCATION EMPLOYMENT RS 2.2 b) Given the changes in technology and market demands, continuous training and (eventual) retraining of the workforce in the forestry sector is required
37. EDUCATION EMPLOYMENT RS 2.5 Employed professional and qualified workforce.
38. EDUCATION EMPLOYMENT SI 2.9 Prioritize Personnel Development: Identify and bridge knowledge gaps among employees based on anticipated industry trends over the next 5 to 10 years. Design and implement training programs accordingly. Provide Specialized Training: Instead of generic CSR trainings, offer specific training in areas like carbon footprint relevant to the company's industry. Ensure that such trainings are solution-oriented and address real challenges faced by the company.
39. EDUCATION POLICY SK 2.3 Prioritize education and capacity building to develop skilled workforce capable of driving innovation in bioeconomy

40. EMPLOYMENT HR 2.3 Securing work-life balance in new jobs created in bioeconomy
41. EMPLOYMENT PL 2.4 Transformation and re-skilling of employees in mining (e.g. coal) and related sectors
42. EMPLOYMENT EDUCATION HR 2.5 Promoting bioeconomy jobs with the youth population
43. EMPLOYMENT EDUCATION HR 2.6 Creating reskilling educational programs in bioeconomy as a measure to preventing the unemployment
44. EMPLOYMENT POLICY CZ 2.6 New jobs in the circular bioeconomy should be supported through financial incentives and similar economic instruments from public budgets.
45. EMPLOYMENT RO 2.4 create green jobs, special attention to rural communities

46. EQUITY PL 2.5 Ensuring women have equal opportunities to build a shared, sustainable future (Women must have equal access to both work on the factory floor and in high-tech laboratories)
47. EQUITY RS 2.1 a)to ensure that women and men have equal opportunities in the forestry sector
48. EQUITY EDUCATION GR 2.2 Encourage the participation of women in decision-making processes related to bioeconomy initiatives.
49. EQUITY EDUCATION SI 2.1 Youth Engagement: Ensure the inclusion of young minds in the bioeconomy transformation. New Opportunities for the Youth: Launch internships, scholarships, and innovation competitions. Collaborate with universities to provide hands-on experiences in the sector.
50. EQUITY EMPLOYMENT HR 2.1 Securing equal job opportunities for women
51. EQUITY POLICY GR 2.1 Implement policies and programs that address gender biases and stereotypes, fostering a supportive environment for women in bioeconomy-related fields.
52. EQUITY POLICY GR 2. 5 Ensure that bioeconomy initiatives do not disproportionately affect vulnerable communities and work towards inclusive development.
53. EQUITY POLICY PL 2.6 Tackling demographic and social changes related to an ageing population, internal and external migration of young people - affecting various aspects of economic and social development in the countryside and slowdown in generational change in sectors of bioeconomy e.g. agriculture or fisheries (Need to Adopt National/Regional Bioeconomy Strategies - Negative Effects of Ageing and Demographics)
54. POLICY CZ 2.5 Food security should always take precedence over energy security in decision-making.
55. POLICY GR 2.6 Promote flexible work arrangements and family-friendly policies within the bioeconomy sector, ensuring a healthy work-life balance
56. POLICY SI 2.5 Simplify Legal and Permit Procedures: Expedite the process for companies to obtain essential environmental and spatial permits, reducing bureaucratic delays that hinder the bioeconomy projects.
57. POLICY SI 2.6 Address Political Bottlenecks: Identify and rectify the lack of political support that maintains the agricultural status quo, hindering progress in sustainable practices.
58. POLICY EMPLOYMENT HU 2.5 As stated in the Concept Paper (BIOEAST) for Hungary, one of the priorities of the future bioeconomy strategy in Hungary will be "food first and healthy diet", which will support local organic farming while raising awareness on healthy diet in society.

3. finance and investment

1. COOPERATION BG 3.2 Cooperation with private investors to build and maintain a stable logistics and supply chain infrastructure for smooth movement of bio-based products.
2. COOPERATION CZ 3.2 Financial support for cooperation and networking among key stakeholders
3. COOPERATION GR 3.2 Foster collaboration between the government, private sector, and research institutions through PPPs to leverage shared resources, knowledge, and funding for bioeconomy projects.
4. COOPERATION GR 3.5 Collaborate with private investors to build and maintain a robust logistics and supply chain infrastructure for the seamless movement of bio-based products.
5. COOPERATION GR 3.6 Collaborate with financial institutions and regulatory bodies to establish clear standards for green finance, ensuring transparency and accountability in bioeconomy-related investments.
6. COOPERATION HU 3.1 Attract private and public investment to support bio-based industries and research.
7. COOPERATION HU 3.2 Foster public-private partnerships to finance and scale bioeconomy ventures, particularly in promoting sectors like bioenergy, sustainable agriculture, and forestry.
8. COOPERATION PL 3.6 Building and leveraging the first of its kind bioeconomy entrepreneurship ecosystem, supported by public and private funding institutions
9. COOPERATION RO 3.2 create synergies and complementarities between funds
10. COOPERATION RO 3.3 Integrated projects: private and public stakeholders (e.g. developing integrated plans in rural communities taking into consideration social-economic-environment dimensions)
11. COOPERATION RO 3.4 Medium and large-scale business cases supporting private BE infrastructure;
12. COOPERATION SI 3.2 Create Open-Access Pilot Infrastructure: Establish a conducive environment for pilot infrastructures, ensuring broader access beyond private companies and addressing commercial limitations. Launch pilot biorefineries, especially in sectors like bakery, bread-making, and alternative proteins.
13. COOPERATION SI 3.3 Private Funding: Invite Investors to a discussion on how to promote a pipeline of investable projects in collaboration: ALFI, Peččnik, Pipistrel, SPS.
14. COOPERATION SI 3.4 Involvement of Banks: Invite banks with green bonds on a discussion on how to promote a pipeline of bankable projects in collaboration: SID, NLB, Intesa Sanpaolo Bank.
15. COOPERATION SI 3.6 Partnerships: Facilitate connections between businesses looking for partnerships.
16. COOPERATION SI 3.8 Promote Decisive Action: Redirect focus from trivial debates to initiate bold and impactful steps that will elevate the circular bioeconomy in Slovenia, prioritizing substantial investments in the sector.
17. COOPERATION SK 3.2 Foster collaboration between public and private sectors through partnerships that leverage both financial and non-financial resources for the advancement of bioeconomy
18. EDUCATION CZ 3.4 Financial support for formal and non-formal education in the field of circular bioeconomy
19. EDUCATION SI 3.5 Enhance Investment Knowledge: Promote awareness and training on modern investment collaboration models like Joint Ventures. Reevaluate and clarify existing

partnership stipulations to ensure transparency, scalability, and increased motivation from the private sector.

20. EDUCATION COOPERATION SI 3.9 Real Life Cases: Provide Case studies on investments for deploying circular bioeconomy value chains with business cases.
21. FUNDING BG 3.1 Objective(s) - 1. Invest in infrastructure development, including research facilities, biorefineries and waste management systems, to grow bioeconomy sectors.
22. FUNDING BG 3.3. Financing in the development and use of high-tech productions based on innovations and the latest achievements of science.
23. FUNDING CZ. 3.7 The bioeconomy offers a comprehensive environmental solution. A possible source of funding for the circular bioeconomy could be the transfer of funds from subsidy titles that only address part of the issue.
24. FUNDING PL 3.1 Increasing access to finance and advancing the level of public-private endeavor in pooling resources for R&I
25. FUNDING PL 3.3 Increasing the investment in R&D in bioeconomy sectors
26. FUNDING PL 3.5 Strengthening infrastructure for production plants and industrial processes conversion and pilot test areas to increase the level of technological readiness.
27. FUNDING POLICY SK 3.4 Provide financial support and incentives for start-ups and SMEs in bioeconomy
28. FUNDING RO 3.5 Small scale business cases individual projects;
29. POLICY HR 3.3 Defining the objectives within the strategic national documents under which biodiversity projects can be funded
30. POLICY SK 3.1 Adopt financial strategies and tools that prioritize impact, inclusivity and long-term value creation
31. POLICY CZ 3.1 Clear conditions for funding scientific research in circular bioeconomy (appropriate subsidy titles, long-term strategic documents guaranteeing continuity of the bioeconomy as a research topic).
32. POLICY CZ 3.3 Support of commodification of science and research results in the circular bioeconomy.
33. POLICY CZ 3.5 The creation of standards for bio-economic production (sustainability, circularity, etc.) that would make financial aid possible
34. POLICY CZ 3.6 Investment incentives for investors in the circular bioeconomy
35. POLICY GR 3.1 Create a conducive regulatory environment that encourages domestic and foreign investment in bioeconomy sectors, offering incentives such as tax breaks, grants, and subsidies.
36. POLICY GR 3.3 Support venture capital firms and angel investors specializing in green technologies and bioeconomy projects, ensuring a steady flow of funds into promising ventures.
37. POLICY GR 3.4 Invest in infrastructure development, including research facilities, biorefineries, and waste management systems, to support the growth of bioeconomy sectors.
38. POLICY HR 3.1 Securing enough finances, by simplifying the procedures for national co-financing of the research and innovation projects in the area of bioeconomy
39. POLICY HR 3.2 Defining priority projects for investments in the field of bioeconomy to foster further development
40. POLICY HR 3.4 Facilitating private investments into bioeconomy projects
41. POLICY HU 3.3 Implement financial incentives and tax breaks to encourage investment in bioeconomy initiatives, ensuring a return on investment while contributing to societal and environmental well-being.

42. POLICY HU 3.4 Create financial mechanisms and platforms that allow for easy access to funding, especially for startups and small to medium-sized enterprises venturing into bioeconomy innovations.
43. POLICY HU 3.5 Increase transparency of green financing. Incentivize the banking sector about green financing, and diversity their investment to other areas besides energy efficiency
44. POLICY HU 3.6 Support of small-scale, new generations of biorefineries, supporting local markets and finding a balance in the biomass energy mix.
45. POLICY PL 3.2 Accelerating sector's technological modernization - developing integrated resource-based strategies
46. POLICY PL 3.4 Increasing the application for funding under the European Research Area by universities, institutes and companies.
47. POLICY RO 3.6 encourage and initiate new ways of financing, launch innovative funding schemes
48. POLICY RS 3.2 Increasing the export and trade of forest products can provide additional revenues that can be used to finance the Bioeconomy Strategy. This includes the export of wood products, biomass, non-wood forest products, etc.
49. POLICY RS 3.3 Ad b) Where finance should go- projects on bioeconomy and environment- a clear correlation between the existing green credits and incentives and projects that have a bioeconomic character and make the necessary improvements to the environment for investments and financing.
50. POLICY SI 3.1 Highlight the profitability and ROI of new technologies in the bioeconomy sector.
51. POLICY SI 3.7 Revamp Funding Schemes: Overhaul the national public funding structure to be more responsive to current challenges. Streamline the bureaucratic processes, foster inter-ministerial collaboration, and increase project greenlighting to motivate and involve the private sector more effectively.
52. POLICY SK 3.3 Increase the income of farmers and foresters by increasing the added value of bio-based products, services and technologies, creating new sources of income not only for farmers but for the entire value chains and for the country
53. POLICY COOPERATION RS 3.1 ad a) Allocate funds from the state, provincial and local self-government budgets to support the implementation of the Bioeconomy Strategy Use of international funds (e.g. IPA, Interreg) and EU funds (e.g. Horizon Europe) The private sector, including forestry companies and the wood industry, can participate in the financing
54. POLICY EDUCATION RO 3.1 Awareness raising about financing mechanisms. Support and comprehensive technical assistance related to financial instruments.

4. environment

1. BETTER USE OF NATURAL RESOURCES RO 4.3 circular and value added use of biomass
2. BETTER USE OF NATURAL RESOURCES RO 4.1 efficient management of natural resources
3. BETTER USE OF NATURAL RESOURCES RS 4.3 Circular use of wood products
4. BETTER USE OF NATURAL RESOURCES SI 4.2 Waste Minimization: Achieve near-zero waste in agricultural and industrial processes through biorefineries and circular value chains.
5. BETTER USE OF NATURAL RESOURCES SI 4.3 Energy Self-Sufficiency: Attain a higher degree of energy self-sufficiency using renewable sources such as biomass.
6. BETTER USE OF NATURAL RESOURCES ECONOMIC RO 4.2 focus on the cascading use of biomass to provide products with high added value
7. BETTER USE OF NATURAL RESOURCES GLOBAL CHANGE ECONOMIC SK 4.1 Foster practices that contribute to implementing the principles of circular bioeconomy with the aim of minimizing waste, reducing greenhouse gas emissions and designing added-value products with a lifecycle approach
8. BETTER USE OF NATURAL RESOURCES GLOBAL CHANGE EDUCATION HR 4.2 Promoting bio-based and nature-based solutions in bioeconomy projects
9. BETTER USE OF NATURAL RESOURCES GLOBAL CHANGE SK 4.2 Mitigate environmental challenges through innovative technologies and responsible resource management, while harmonizing environmental, economic and social balance
10. BETTER USE OF NATURAL RESOURCES GLOBAL CHANGE NATURE PROTECTION RS 4.6 Pollution reduction, waste management, nature protection
11. BETTER USE OF NATURAL RESOURCES GLOBAL CHANGE RS 4.4 Forest biomass as a substitute for fossil fuels
12. BETTER USE OF NATURAL RESOURCES POLICY CZ 4.2 Newly established plants should (where possible) be exclusively oriented towards the circular bioeconomy.
13. ECONOMIC BG4.3 Economics in favor of consumers
14. ECONOMIC RS 4.1 Development of sustainable products and innovations
15. ECONOMIC RS 4.2 Development of rural areas
16. ECONOMIC EDUCATION HR 4.3 Encouraging production of recycable materials packaging and bio-based plastics
17. ECONOMIC POLICY BG 4.1 - 1. Green and competitive economy;
18. ECONOMIC POLICY PL 4.3 Accelerating the energy transition while taking care of biodiversity, natural resources (e.g. water resources)
19. ECONOMIC BETTER USE OF NATURAL RESOURCES HU 4.1 To progressively shift towards renewable and sustainable resources (for instance, utilizing the significant biomethane potential of Hungary), minimizing reliance on non-renewable sources (gaining energy independence from other countries in the meantime) and mitigating environmental degradation while maintaining or ideally increasing land productivity (especially relevant for South-East Region of Hungary where most of our intensive **agriculture** takes place and where flooding and droughts are commonplace along with general soil degradation).

20. EDUCATION GR 4.5 Raise public awareness about the benefits and challenges of the bioeconomy.
21. EDUCATION RS 4.8 Raising awareness of the importance of bioeconomy and circular economy
EDUCATION BETTER USE OF NATURAL RESOURCES GR 4.1 Promote sustainable management of natural resources, including **agricultural** residues, forestry products, and marine resources.
22. EDUCATION ECONOMIC GR 4.3 Enhance **agricultural** practices for higher yields and sustainability.
23. GLOBAL CHANGE GR 4.4 Mitigate the impact of **climate** change through bio-based solutions.
24. GLOBAL CHANGE HU 4.3 To contribute to mitigating **climate** change effects and adapting to the changing **climate** scenario.
25. GLOBAL CHANGE RS 4.5 Climate change management (eg forests play a key role in absorbing carbon from the atmosphere, reducing the greenhouse effect)
26. GLOBAL CHANGE SI 4.1 Reduction of Carbon Footprint: Achieve a significant reduction in carbon emissions through the adoption of bioeconomy principles.
27. GLOBAL CHANGE SI 4.4 Climate Resilience: Develop climate-resilient agricultural practices, safeguarding food security and rural livelihoods
28. GLOBAL CHANGE RESEARCH 4.4 Development of the 4th generation biogas and biomethane sector, taking into account emission reduction and climate change potentials
29. NATURE CONSERVATION CZ 4.1 Respect for the nature and integrity of the ecosystems to be used for biomass. Identification of (already protected) ecosystems that will not be used for commercial biomass extraction.
30. NATURE CONSERVATION HU 4.2 To safeguard and restore ecosystems (especially wetlands in the case of Hungary), fostering biodiversity and preserving natural habitats while taking advantage of nature-based solutions.
31. NATURE CONSERVATION RS 4.7 Revitalization and preservation of natural resources - soil, water, etc., restoration of ecosystems, sustainability of biodiversity ...
32. POLICY CZ 4.3 Implementation of the circular bioeconomy serves to meet most of the SDGs
33. POLICY GR 4.6 Develop a robust regulatory framework to ensure the sustainability and ethical practices within the bioeconomy.
34. POLICY HR 4.1 Supporting the projects in bioeconomy that lead to accomplishing climate neutrality and enhancing energy efficiency
35. POLICY HR 4.5 Implementing green procurement procedures in bioeconomy
36. POLICY BETTER USE OF NATURAL RESOURCES BG 4.2. Less waste, more resources;
37. POLICY BETTER USE OF NATURAL RESOURCES HU 4.4 To foster circular bioeconomy by minimizing waste and maximizing resource use efficiency.
38. POLICY BETTER USE OF NATURAL RESOURCES PL 4.6 Sustainable water management, including ensuring access to clean water for the public and the economy and achieving good water status
39. POLICY ECONOMIC GR 4.2 Build a circular economy model, minimizing waste and maximizing resource efficiency.
40. POLICY GLOBAL CHANGE PL 4.1 Transforming the national economy into a resource-efficient, green and competitive low-carbon economy
41. POLICY GLOBAL CHANGE NATURE CONSERVATION CZ 4.4 Consistent implementation of circular bioeconomy principles will contribute to the reduction of negative externalities (especially global climate change)
42. POLICY GLOBAL CHANGE PL 4.5 Increasing the share of renewable energy at national level

- 43. POLICY RESEARCH PL 4.2 Developing fossil fuel substitution pathways
- 44. POLICY NATURE CONSERVATION RS 4.9 Reducing pressure on protected natural areas

